

CLASSIFYING LOW-VOLUME NON-COMPACT ARITHMETIC HYPERBOLIC 3-ORBIFOLDS

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ABSTRACT. We prove results about minimal orbifolds and their covers in arithmetic commensurability classes, with a view toward practical classification problems. As an application, we classify the non-compact hyperbolic 3-orbifolds with volume at most v^* , the minimal volume of non-compact, non-arithmetic hyperbolic 3-orbifolds identified recently by Drewitz–Kellerhals.

1. INTRODUCTION

This paper addresses aspects of the problem of enumerating all members of a given commensurability class $\mathcal{C}$ of arithmetic hyperbolic orbifolds, with one representative per isometry class, having volume at most a given bound. Here two orbifolds are *commensurable* if they have a common finite-degree cover; the *commensurability class* of an orbifold is the set of all orbifolds commensurable to it.

A venerable tool from computational group theory, the *low-index subgroups algorithm*, allows us to enumerate all covers of a given hyperbolic orbifold P up to a given (reasonable) degree bound, provided that we have a presentation for the orbifold fundamental group of P . The basic idea is to apply this algorithm to a member or members of $\mathcal{C}$ that are minimal with respect to the covering relation. Issues arise from the fact that as an *arithmetic* commensurability class, $\mathcal{C}$ has infinitely many minimal such members, multiple of which may have volume at most the given bound v . We can conceive two approaches to dealing with this:

- (1) List all covers having volume at most v of each relevant minimal orbifold in $\mathcal{C}$, then merge the resulting lists (eliminating isometric duplicates).
- (2) Fix a single minimal orbifold Q_0 ; list all of its covers of volume up to an appropriate multiple of v ; and encode each member P of $\mathcal{C}$ that does not cover Q_0 in terms of a common cover $\tilde{P}$ of both P and Q_0 from that list plus extra data.

Here we use approach (1), leveraging existing descriptions of low-volume minimal orbifolds, to locate the smallest non-compact arithmetic hyperbolic 3-orbifolds within their commensurability classes. Below for $d \in \mathbb{N}$ square-free, O_d is the ring of integers of $\mathbb{Q}(\sqrt{-d})$; $\mathrm{GL}_2(O_d)$ is the group of 2×2 matrices with entries in O_d and determinants in its group of units O_d^* ; and $\mathrm{PGL}_2(O_d)$ is the quotient of $\mathrm{GL}_2(O_d)$ by its subgroup of diagonal matrices.

By work of Borel [3] reviewed in Section 3, the set of minimal orientable orbifolds commensurable to $\mathbb{H}^3/\mathrm{PGL}_2(O_d)$ corresponds bijectively to the set of primes and products of distinct prime ideals in O_d . For $d = 1$ and 3, the cases of interest here, we recall that O_d is a *PID*: its ideals have the form pO_d for $p \in O_d$.

Theorem 1.1. *Let $v_7 \approx 0.44445$ be the volume of $\mathbb{H}^3/\mathrm{PGL}_2(O_7)$. A complete, non-compact arithmetic hyperbolic 3-orbifold with volume less than $v_7/2 \approx 0.222225$ is commensurable to exactly one of $Q_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_3)$ or $R_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_1)$. Exactly nineteen members of the commensurability class $\mathcal{C}_3$ of Q_0 have volume less than $v_7/2$. In order of increasing volume, the orientable such members are:*

- *With volume $\frac{v_0}{12}$, where $v_0 \approx 1.01494146$ is the volume of a regular ideal tetrahedron in $\mathbb{H}^3$: Q_0 .*
- *With volume $v_0/6$: the unique double cover O^2 of Q_0 , and the minimal orientable orbifold $Q_{\sqrt{-3}} = \mathbb{H}^3/\Gamma_{\sqrt{-3}}$ in $\mathcal{C}_3$ associated to the prime $\sqrt{-3}O_3$.*
- *With volume $5v_0/24$: the minimal orientable orbifold $Q_2 = \mathbb{H}^3/\Gamma_2$ in $\mathcal{C}_3$ associated to the prime $2O_3$.*

Each of Q_0 , $Q_{\sqrt{-3}}$, and Q_2 double-covers a unique non-orientable orbifold having half its volume, denoted as $\bar{Q}_0$, $\bar{Q}_{\sqrt{-3}}$, and $\bar{Q}_2$, respectively. There are exactly two non-orientable orbifolds double-covered by O^2 . In order of increasing volume, the remaining non-orientable members of $\mathcal{C}_3$ are:

- *With volume $v_0/8$: the unique non-orientable orbifold $\bar{O}^3$ double covered by the unique degree-3 cover O^3 of Q_0 .*
- *With volume $v_0/6$: exactly two orbifolds double-covered by the unique degree-4 cover O^4 of Q_0 .*
- *With volume $5v_0/24$: a total of six distinct orbifolds respectively double-covered by the double covers O^5 , P_2^1 and P_2^2 of Q_2 , where O_5 also covers Q_0 ; and the unique orbifold $\bar{Q}_{2\sqrt{-3}}$ double covered by the minimal orientable orbifold $Q_{2\sqrt{-3}}$ associated to the product of primes $2\sqrt{-3}O_3$.*

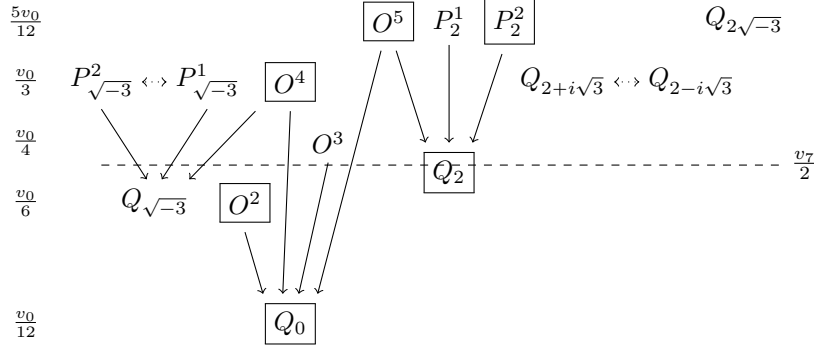
Exactly nine members of the commensurability class $\mathcal{C}_1$ of R_0 have volume less than $v_7/2$. In order of increasing volume, these are classified up to isometry as follows:

- *With volume $v_1/12$, where $v_1 \approx 0.91596559$ is Catalan's constant, the volume of an ideal tetrahedron in $\mathbb{H}^3$ with dihedral angles $\frac{\pi}{2}$, $\frac{\pi}{4}$, and $\frac{\pi}{4}$: the unique non-orientable orbifold $\bar{R}_0$ double-covered by R_0 .*
- *With volume $\frac{v_1}{8}$: the unique non-orientable orbifold $\bar{R}_{1+i}$ double-covered by the minimal orbifold R_{1+i} in $\mathcal{C}_1$ associated to the prime $(1+i)O_1$.*
- *With volume $\frac{v_1}{6}$: R_0 , or one of six non-orientable orbifolds double-covered by the three double covers of R_0 —two each.*

Section 2 describes each orbifold of Theorem 1.1; Section 4 gives its proof. This builds on existing work: the reduction to the commensurability classes $\mathcal{C}_1$ and $\mathcal{C}_3$, in Lemma 4.1, is adapted from [13, Th. 11.7.2], and we use explicit descriptions of the orbifolds O_2 , Q_0 , R_0 , Q_2 , R_2 , and $Q_{\sqrt{-3}}$ given (by other names) in [16].

Theorem 1.1's classification of members of $\mathcal{C}_3$ follows from an analogous classification of its *orientable* members having volume less than v_7 , which is summarized in Figure 1. The pairs $P_{\sqrt{-3}}^1$ and $P_{\sqrt{-3}}^2$, and $Q_{2\pm i\sqrt{3}}$, of the Figure are each orientation-reversing isometric via isometries induced by complex conjugation. Those boxed in the Figure are *tetrahedral*, quotients of $\mathbb{H}^3$ by the orientation-preserving subgroups of groups generated by reflections in the faces of certain tetrahedra; cf. [13, §13.2.1].

Combining Theorem 1.1 with the recent work [9] of Drewitz–Kellerhals identifying the smallest *non-arithmetic* non-compact hyperbolic 3-orbifold yields a classification of all complete, non-compact hyperbolic 3-orbifolds having at most its volume v_* , upon noting that $5v_0/24 \approx 0.211446 > v_*$, while $v_0/6 \approx 0.169156 < v_*$.


 FIGURE 1. Orientable orbifolds in $\mathcal{C}_3$ having volume $\leq v_7$.

Corollary 1.2. *A complete hyperbolic 3-orbifold of volume at most $v_* \approx 0.171502$, the minimal volume of non-compact, non-arithmetic hyperbolic 3-orbifolds from [9], is either the orbifold identified there, one of the nine members of $\mathcal{C}_1$ listed in Theorem 1.1, or one of the eleven members of $\mathcal{C}_3$ having volume less than $5v_0/24$.*

We remark that C. Adams classified all non-compact orientable hyperbolic orbifolds having volume less than $v_0/4 \approx 0.253735$ and, using this, all non-compact orbifolds with volume less than $v_0/8 \approx 0.126867$, in [1]. (The smallest of these was previously found by Meyerhoff [15].) Eleven—all six non-orientable, and five orientable—have volume less than v^* . The remaining arithmetic orbifolds of Corollary 1.2 arise in [9] as cases in the proof of its main theorem.

In this sense, Theorem 1.1 above performs a similar function for [9] as Neumann–Reid’s paper [16] did for [1], divining relationships between and additional arithmetic information about the examples discovered.

We now outline the structure of the paper. Section 2 recalls the descriptions from [16] of the six smallest orientable orbifolds of Theorem 1.1, and builds from these to describe the others. Section 3 reviews Borel’s results on minimal orbifolds in arithmetic commensurability classes and applies them to $\mathcal{C}_3$, explicitly listing all those of small volume in Theorem 3.4.

The results of Section 3 serve both the present paper and a forthcoming one [8], where we will give a census of all orientable 3-orbifolds in $\mathcal{C}_3$ having volume no greater than the figure-eight knot complement (which itself belongs to $\mathcal{C}_3$). Theorem 3.4 thus lists all minimal orbifolds of $\mathcal{C}_3$ up to its volume of $2v_0 \approx 2.0298$. For most of these we do not know of polyhedral decompositions from the literature, so in [8] we use approach (2) above to describe their covers in terms of those of Q_0 .

Section 3 supplies the underpinnings for this approach: Proposition 3.1 describes, and Definition 3.2 defines, the unique cover $\tilde{Q}_\pi$ of smallest degree common to Q_0 and another minimal orientable orbifold Q_π . By Prop. 3.1, the cover $\tilde{Q}_\pi \rightarrow Q_\pi$ is normal. In [8] we will understand an arbitrary cover $P \rightarrow Q_\pi$ in terms of a cover $\tilde{P}$ of $\tilde{Q}_\pi$ and lifts of covering transformations of $\tilde{Q}_\pi \rightarrow Q_\pi$. A key additional tool for [8] is a *canonical decomposition* of covers of $Q_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_3)$ that allows us to completely understand the isometries between any pair of them. This was established in the third author’s PhD thesis [11].

1.1. The basics of hyperbolic orbifolds. Here we briefly lay out a sequence of definitions and facts that we will take for granted in the rest of the paper. We will work in the upper half-space model $\mathbb{C} \times (0, \infty)$ for $\mathbb{H}^3$, whose full isometry group has the form $\mathrm{PSL}_2(\mathbb{C}) \rtimes \langle r \rangle$, where r is the reflection in the totally geodesic plane $\mathbb{R} \times (0, \infty)$. Here $\mathrm{PSL}_2(\mathbb{C})$ is the group of orientation-preserving isometries, acting extending its action on $\mathbb{C} \cup \{\infty\}$ by Möbius transformations. The action of r extends $z \mapsto \bar{z}$, and it conjugates $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{PSL}_2(\mathbb{C})$ to $\begin{pmatrix} \bar{a} & \bar{b} \\ \bar{c} & \bar{d} \end{pmatrix}$.

The definition of an orbifold is widely attributed to Satake [18], but the ur-reference for the below is Thurston's notes [19, Ch. 13]. For expansion on and/or variations of Thurston's approach, see eg. Boileau–Maillot–Porti [2, Ch. 2], Bridson–Haefliger [4, Ch. III.G], Choi [6], and Ratcliffe [17, Ch. 13].

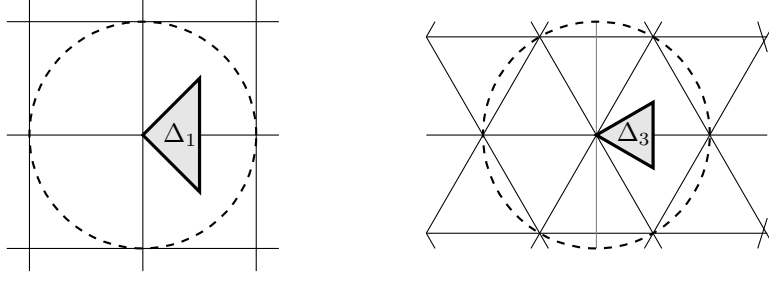
- A *complete hyperbolic 3-orbifold* is a quotient $O = \mathbb{H}^3/\Gamma$ of three-dimensional hyperbolic space by the (left-)action of a discrete group Γ of isometries, its *orbifold fundamental group*. It is *orientable* if every element of Γ preserves orientation; otherwise it is *non-orientable*.
- The *volume* $\mathrm{vol}(O)$ of O is that of a fundamental domain for Γ in $\mathbb{H}^3$.
- The *singular locus* $\Sigma(O)$ of O is the projection of the set of fixed points in $\mathbb{H}^3$ of the non-trivial elements of Γ . The *isotropy group* of $u \in \Sigma(O)$ is the isomorphism class of the stabilizer in Γ of any $\tilde{u} \in \mathbb{H}^3$ projecting to u .
- For a group Γ_0 and isometry g such that $g\Gamma_0g^{-1} \leq \Gamma$, the map $\mathbb{H}^3/\Gamma_0 \rightarrow O$ induced by g is an *orbifold cover* of O , and every orbifold cover arises this way. Its *degree* is the index $[\Gamma : g\Gamma_0g^{-1}]$. An *isometry* is a degree-1 cover.
- An isometry between orientable orbifolds is *orientation-preserving* or *reversing* according to the nature of an isometry of $\mathbb{H}^3$ that induces it.
- Covers of O corresponding to subgroups Γ_0 and Γ_1 of Γ are *equivalent* if and only if Γ_0 is conjugate to Γ_1 within Γ . Elements of the normalizer of Γ_0 in Γ induce *covering transformations* of the corresponding cover $\mathbb{H}^3/\Gamma_0$. A *normal* cover corresponds to a normal subgroup.

We will use the *Poincaré polyhedron theorem* as laid out in [17, Th. 13.5.3], the first complete proof of which is commonly attributed to Maskit [14]. This result gives conditions ensuring that a collection of face-pairings of a hyperbolic polyhedron P generates a discrete group Γ having P as a fundamental domain, and if so, provides a presentation for Γ .

2. A FIRST LOOK AT THE ORBIFOLDS WE AIM TO CLASSIFY

The present section explicitly describes the orbifolds of Theorem 1.1. Our work here builds on Neumann–Reid's descriptions, in [16], of the orbifolds discovered by Adams [1]. We re-describe these orbifolds, extract presentations for their orbifold fundamental groups; and, using this, we enumerate their low-index subgroups relevant to the current classification and further describe the corresponding covers.

We will use the following notation from [16]: denote by $T(\alpha_1, \alpha_2, \alpha_3; \beta_1, \beta_2, \beta_3)$ the tetrahedron in $\mathbb{H}^3$ (possibly with some ideal vertices) having dihedral angles π/α_i , $i = 1, 2, 3$ around the edges of one face, and angle π/β_i around the edge opposite (ie. not sharing an endpoint with) the π/α_i edge, for each i . The *tetrahedral rotation group* in $T(\alpha_1, \alpha_2, \alpha_3; \beta_1, \beta_2, \beta_3)$ is the index-two orientation-preserving subgroup of the group generated by reflections in the faces of this tetrahedron.


 FIGURE 2. Fundamental domains for the $\mathrm{PGL}_2(O_d)$ -actions, $d = 1, 3$.

2.1. The smallest members of $\mathcal{C}_1$ and $\mathcal{C}_3$. The two smallest orientable cusped orbifolds are $Q_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_3)$ and $R_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_1)$, the latter of which is termed Q_1 in [16]. Here for $d = 1, 3$, O_d is the ring of integers of $\mathbb{Q}(\sqrt{-d})$, and $\mathrm{PGL}_2(O_d) = \mathrm{GL}_2(O_d)/\{(\begin{smallmatrix} x & 0 \\ 0 & x \end{smallmatrix}) \mid x \in O_d^*\}$, where O_d^* is the group of units in O_d and

$$\mathrm{GL}_2(O_d) = \{(\begin{smallmatrix} x & y \\ z & w \end{smallmatrix}) \mid x, y, z, w \in O_d, xw - yz \in O_d^*\}.$$

The group O_1^* has generator i and order four, and O_3^* is generated by $\zeta = \frac{1}{2}(1 + \sqrt{-3})$, with order six.

As mentioned in [16] and described in more detail by Hatcher [12], for $d = 1, 3$ the $\mathrm{PGL}_2(O_d)$ are the orientation-preserving symmetry groups of tessellations $\mathcal{T}_d$ of $\mathbb{H}^3$ by regular ideal tetrahedra and octahedra, respectively. The ideal vertices of $\mathcal{T}_d$ are at points of $\mathbb{Q}(\sqrt{-d}) \cup \{\infty\}$ in the upper half-space model, with vertices α/β and γ/δ ($\alpha, \beta, \gamma, \delta \in O_d$) joined by an edge if and only if $\alpha\delta - \beta\gamma = O_d^*$. In particular, the cells that have a vertex at ∞ have their other ideal endpoints in O_d , respectively the lattices in $\mathbb{C}$ spanned by 1 and i , or by 1 and ζ . Viewed from above, these collections of cells look like tilings of $\mathbb{C}$ by squares or equilateral triangles, in each case having an edge joining 0 to 1.

For either $d = 1$ or 3, after dividing the cells of $\mathcal{T}_d$ barycentrically into tetrahedra, a fundamental domain for the action of $\mathrm{PGL}_2(O_d)$ on $\mathcal{T}_d$ consists of the union of one of the resulting tetrahedra with its mirror image across a face. Figure 2 gives the view from above of one such choice T_d of fundamental domain.

Definition 2.1. Let Δ_1 (respectively, Δ_3) be the triangle in $\mathbb{C}$ with vertices at 0 and $\frac{1}{2}(1 \pm i)$ (resp. $\frac{1}{2}(1 \pm i/\sqrt{3})$), and let $T_d = (\Delta_d \times (0, \infty)) \cap (\mathbb{H}^3 - \mathbb{B}^3)$, where $\mathbb{B}^3$ is the open unit ball centered at the origin in the Euclidean metric on $\mathbb{C} \times \mathbb{R}$.

Let v be the face $T_d \cap \overline{\mathbb{B}^3}$ of T_d , where $\overline{\mathbb{B}^3}$ is the closed unit ball. The other three faces are contained in $\partial\Delta_d \times (0, \infty)$; let e lie over the line segment from $\frac{1+i}{2}$ to $\frac{1-i}{2}$ (respectively, from $\frac{1}{2}(1 - i/\sqrt{3})$ to $\frac{1}{2}(1 + i/\sqrt{3})$), f_+ over the segment from 0 to $\frac{1+i}{2}$ (resp. $\frac{1}{2}(1 + i/\sqrt{3})$), and f_- over the segment from 0 to $\frac{1-i}{2}$ (resp. $\frac{1}{2}(1 - i/\sqrt{3})$).

For each d , T_d is divided by the vertical plane over the real axis into the union of two barycentric subdivision tetrahedra for $\mathcal{T}_d$, mirror images of each other across their shared a face in this plane. In the notation from [16] described above, these tetrahedra are respectively $T(3, 2, 2; 4, 2, 4)$ ($d = 1$) and $T(3, 2, 2; 6, 2, 3)$ ($d = 3$).

A single such tetrahedron is a fundamental domain for the full symmetry group of the tiling $\mathcal{T}_d$, generated by the reflections in its faces, which determines the orbifolds that we call $\bar{R}_0$ and $\bar{Q}_0$ in the respective cases of Theorem 1.1. The subgroups

$\mathrm{PGL}_2(O_d)$ consisting of orientation-preserving isometries are thus tetrahedral rotation groups, generated by three rotations—products of said reflections—acting as face-pairings of T_d :

$$\begin{aligned} (1) \quad (d=1) \quad & \rho_f = \begin{bmatrix} i & 0 \\ 0 & 1 \end{bmatrix} & \rho_e = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix} & \rho_v = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ (2) \quad (d=3) \quad & \rho_f = \begin{bmatrix} \zeta & 0 \\ 0 & 1 \end{bmatrix} & \rho_e = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix} & \rho_v = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{aligned}$$

Here, square brackets around a matrix denote its equivalence class in $\mathrm{PGL}_2(O_3)$.

The isometry ρ_f rotates counterclockwise about the axis from 0 to ∞ , which is fixed, and taking the vertical face f_- to f_+ . The other two each identify a face of T_d to itself: ρ_e , which rotates by an angle of π about an axis from $1/2$ to ∞ , acts by reflection on the vertical face e ; and ρ_v , which rotates by π around an axis through ± 1 , acts by reflection on the face v in $\partial \overline{\mathbb{B}}^3$.

By the Poincaré polyhedron theorem (see eg. [17, Ch. 11]), the conjugacy classes of elliptic elements in $\mathrm{PGL}_2(O_d)$ correspond to *edge cycles* of the fundamental domain T_d , orbits of its edges under the side pairings. Each of the generators above determines such a cycle consisting of a single edge contained in its axis. (For the generators ρ_v and ρ_e , these “edges” bisect faces of T_d and we treat them as having dihedral angle π .) There are three more edge cycles of T_d , each with a representative fixed by one of the products:

- (1) $\rho_f \rho_e$, rotating around the edge $f_+ \cap e$ with an endpoint at ∞ ;
- (2) $\rho_f \rho_v$, rotating around the edge $f_+ \cap v$;
- (3) $\rho_e \rho_v$, rotating around $e \cap v$, which we regard as a union of two edges.

Each elliptic above has two edges in its associated edge cycle. Using their dihedral angles and the Poincaré polyhedron theorem, we thus obtain the presentations:

$$\begin{aligned} (3) \quad \mathrm{PGL}_2(O_1) &\cong \langle \rho_f, \rho_e, \rho_v \mid \rho_f^4 = \rho_e^2 = \rho_v^2 = (\rho_f \rho_e)^4 = (\rho_f \rho_v)^2 = (\rho_e \rho_v)^3 = 1 \rangle \\ (4) \quad \mathrm{PGL}_2(O_3) &\cong \langle \rho_f, \rho_e, \rho_v \mid \rho_f^6 = \rho_e^2 = \rho_v^2 = (\rho_f \rho_e)^3 = (\rho_f \rho_v)^2 = (\rho_e \rho_v)^3 = 1 \rangle \end{aligned}$$

The Fact below can be proved directly:

Fact 2.2. The following holds for the tetrahedron T_d of Definition 2.1 and its faces:

- (1) T_d has no orientation-preserving self-isometries.
- (2) The reflection r in the plane $\mathcal{H}$ over $\mathbb{R}$, which acts on $\mathbb{C}$ by complex conjugation, is the sole orientation-reversing self-isometry of T .
- (3) The faces e , v , and f_+ of T_d are pairwise non-isometric, and f_+ has no non-trivial self-isometry.
- (4) The restriction of r to each of v and e is its unique non-trivial self-isometry, and r gives the unique isometry from f_+ to f_- . These isometries reverse the boundary orientation the faces inherit from T_d .

The isometries of Fact 2.2(4) are the restrictions to faces of the maps ρ_v , ρ_e , and ρ_f defined in (2). The reflection r of Fact 2.2(2) normalizes $\mathrm{PGL}_2(O_3)$ in the full isometry group of $\mathbb{H}^3$, conjugating ρ_f to ρ_f^{-1} and ρ_v and ρ_e to themselves. The group generated by $\mathrm{PGL}_2(O_3)$ and this reflection is therefore a degree-two extension of $\mathrm{PGL}_2(O_3)$, acting on $\mathbb{H}^3$ with quotient the reflection orbifold in the faces of T_0 .

2.2. Low-index subgroups of $\mathrm{PGL}_2(O_d)$, $d = 1, 3$. Using the presentations (3) and (4), and a tool such as the `low_index` Python module [7], we can obtain permutation representations corresponding to their low-index subgroups up to conjugacy.

Non-trivial index-two subgroups of $\mathrm{PGL}_2(O_1)$: $[[0, 1], [1, 0], [1, 0]]$,
 $[[1, 0], [0, 1], [0, 1]]$, and $[[1, 0], [1, 0], [1, 0]]$

Non-trivial subgroups of $\mathrm{PGL}_2(O_3)$ up to index five:

- index-two: $[[1, 0], [1, 0], [1, 0]]$
- index-three: $[[0, 2, 1], [1, 0, 2], [0, 2, 1]]$
- index-four: $[[0, 1, 3, 2], [0, 2, 1, 3], [1, 0, 2, 3]]$
- index-five: $[[0, 1, 3, 4, 2], [0, 2, 1, 4, 3], [1, 0, 2, 4, 3]]$

Each list given above prescribes a right-representation (ie. reversing the order of multiplication) to the symmetric group S_n , where n is the length of each sub-list, by encoding the images of the generators ρ_f, ρ_e, ρ_v in that order. For example, under the representation ϕ of $\mathrm{PGL}_2(O_3)$ encoded by the subgroup labeled “index-four”, we would have $\phi(\rho_f)(0) = 0$, $\phi(\rho_f)(1) = 1$, $\phi(\rho_f)(2) = 3$, and $\phi(\rho_f)(3) = 2$. Each such representation has transitive image, and the subgroup associated to it is recovered as the stabilizer of (say) 0. The representation itself records the action of the full group by right-multiplication on the cosets of this subgroup.

Definition 2.3. For each $i \in \{2, 3, 4, 5\}$, let H^i be the stabilizer of 0 under the transitive right-permutation representation $\mathrm{PGL}_2(O_3) \rightarrow S_i$ given above, and let $O^i = \mathbb{H}^3/H^i$ be the corresponding cover of Q_0 .

For a group G of isometries generated by a set of face-pairings, and a right-permutation representation as described above, the lifted polyhedral decomposition of the associated cover may be recovered from the representation as follows:

Lemma 2.4. *Suppose $G < \mathrm{Isom}(\mathbb{H}^3)$ is generated by a set of face-pairings $\rho_1, \dots, \rho_k$ of a polyhedron $P \subset \mathbb{H}^3$, and for each i let $F_i^\pm$ be faces of P such that $\rho_i(F_i^-) = F_i^+$. Suppose $\phi: G \rightarrow S_n$ is a transitive right-representation, and let H be the stabilizer of 0 under the associated right-action on $\{0, 1, \dots, n-1\}$.*

For any $g_0, \dots, g_{n-1} \in G$ such that $0.g_j = j$ for each j , the union of translates $P_j \doteq g_j(P)$ projects onto $\mathbb{H}^3/H$. The faces of the P_j are identified in $\mathbb{H}^3/H$ as follows: for each $i \in \{1, \dots, k\}$ and $j \in \{0, \dots, n-1\}$, taking $j' = \phi(\rho_i)(j)$, the copy $g_{j'}(F_i^-)$ of F_i^- in $P_{j'}$ is identified to the copy of F_i^+ in P_j by $g_j \rho_i g_{j'}^{-1}$.

Proof. Let $g_0, \dots, g_{n-1}$ be elements of G with the property that for each $j \in \{0, \dots, n-1\}$, $\phi(g_j)(0) = j$. These then comprise a complete set of right coset representatives for H in G , since for any $j' \neq j \in \{0, \dots, n-1\}$,

$$\phi(g_{j'} g_j^{-1})(0) = \phi(g_j)^{-1} \phi(g_{j'})(0) = \phi(g_j)^{-1}(j') \neq 0.$$

It follows that the translates of $P_j \doteq g_j(P)$ belong to different H -orbits: if $h g_j(P) = g_{j'}(P)$ for some $j' \neq j$ and $h \in H$, then $h(P) = g_{j'} g_j^{-1}(P)$, contradicting that $g_{j'} g_j^{-1} \notin H$. Since H has index n in G , it follows that the union of the P_j projects onto $\mathbb{H}^3/H$.

For any $i \in \{1, \dots, k\}$ and $j \in \{0, \dots, n-1\}$, if $\phi(\rho_i)(j) = j'$ then $g_j \rho_i$ is in the same right coset of H as $g_{j'}$, since

$$\phi(g_j \rho_i)(0) = \phi(\rho_i) \phi(g_j)(0) = \phi(\rho_i)(j) = j' = \phi(g_{j'})(0).$$

For this j and j' , $g_j \rho_i g_{j'}^{-1}(g_{j'}(P)) = g_j \rho_i(P)$ intersects $g_j(P)$ along the copy of F_i^- in $g_j \rho_i(P)$ and F_i^+ in $g_j(P)$, since $\rho_i(P) \cap P$ is the face F_i^+ of P and $\rho_i(F_i^-)$ in $\rho_i(P)$. For h such that $g_j \rho_i = h g_{j'}$, we have that $g_j \rho_i(P) = h(P_{j'})$ is in the H -orbit of $P_{j'} = g_{j'}(P)$, so the copy of F_i^- in $P_{j'}$ is identified to the copy of F_i^+ in P_j under the projection to $\mathbb{H}^3/H$. $\square$

As a first application of Lemma 2.4, we show that the double cover of $\mathbb{H}^3/\mathrm{PGL}_2(O_3)$ is tetrahedral.

Lemma 2.5. *For $H^2 < \mathrm{PGL}_2(O_3)$ and the degree-two cover $O^2 = \mathbb{H}^3/H^2$ of $Q_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_3)$ as in Definition 2.3, O^2 is tetrahedral. Specifically, H^2 is the tetrahedral rotation group of T_3 , which has the description $T(2, 2, 3; 3, 3, 3)$ in the notation of [16].*

The reflection r in the plane $\mathcal{H}$ over $\mathbb{R}$ normalizes H^2 and induces a self-isometry of O^2 and a degree-two cover from O^2 to $\mathbb{H}^2/\langle H^2, r \rangle$, an orbifold with a single cusp of type $\mathbb{D}^2 : (3; 3)$ in the taxonomy of [19, Th. 13.3.6].

Remark. Above, “cusp type $\mathbb{D}^2(3; 3)$ ” means that a cusp cross-section is a Euclidean 2-orbifold with underlying space $\mathbb{D}^2$, a single cone point of order 3, and mirrored boundary with a single corner reflector of order 3.

Proof. Recall that H^2 is the stabilizer of 0 under the right-permutation representation $\mathrm{PGL}_2(O_3) \rightarrow S_2$ given by:

$$\rho_f \mapsto [1, 0], \quad \rho_e \mapsto [1, 0], \quad \rho_v \mapsto [1, 0].$$

Using ρ_v as the non-trivial coset representative for H^2 , Lemma 2.4 implies that $T_3 \cup \rho_v(T_3)$ is a fundamental domain for the H^2 -action on $\mathbb{H}^3$. Since T_3 intersects $\rho_v(T_3)$ along its face v from Definition 2.1, the faces of $T_3 \cup \rho_v(T_3)$ consist of the other faces e and $f_{\pm}$ of that Definition, together with their translates under ρ_v .

Since ρ_v has order two and exchanges 0 and 1 under the permutation representation, Lemma 2.4 gives the following face-pairings:

- f_- is paired to $\rho_v(f_+)$ by $\rho_v\rho_f$
- $\rho_v(f_-)$ is paired to f_+ by $\rho_f\rho_v$
- e is paired to $\rho_v(e)$ by $\rho_v\rho_e$

By the Poincaré polyhedron theorem, this is a generating set for H^2 . Since each of $\rho_v\rho_f$, $\rho_f\rho_v$, and $\rho_v\rho_e$ fixes an edge of T_3 contained in its face v of intersection with $\rho_v(T_3)$, we have expressed H^2 as the tetrahedral rotation group in T_3 .

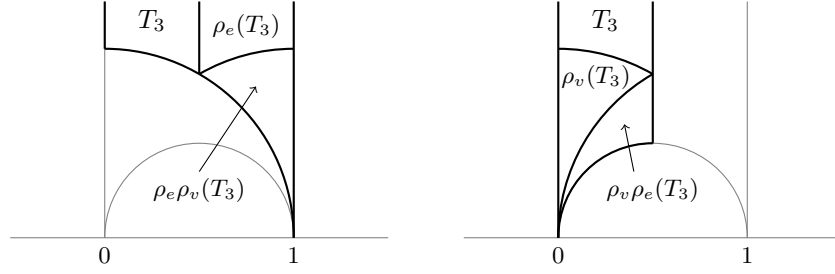
As noted below Fact 2.2, the reflection r conjugates ρ_v and ρ_e to themselves, and it exchanges ρ_f with ρ_f^{-1} . Therefore it commutes with the generator $\rho_v\rho_e$ for H^2 from above, and exchanges the other two. Hence r normalizes H^2 . The fixed locus $\mathcal{H}$ of r bisects $T_3 \cup \rho_v(T_3)$, and we can take one of the resulting halves as a fundamental domain for $\langle H^2, r \rangle$, say the one consisting of points with non-negative imaginary coordinate. A cusp cross-section for $\mathbb{H}^3/\langle H^2, r \rangle$ is assembled by pairing edges of cross-sections at the ideal points 0 and ∞ of this domain, each a triangle with angles $\pi/2$, $\pi/3$, and $\pi/6$. The reflection r fixes each such triangle’s edge opposite the $\pi/3$ -vertex, and the face-pairings ρ_e and $\rho_f\rho_v$ identify the triangles’ other two edges in pairs, yielding $\mathbb{D}^2(3; 3)$. $\square$

Lemma 2.6. *For $H^3 < \mathrm{PGL}_2(O_3)$ and the degree-three cover $O^3 = \mathbb{H}^3/H^2$ of $Q_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_3)$ as in Definition 2.3, the reflection r in the plane $\mathcal{H}$ over $\mathbb{R}$ normalizes H^3 and induces the unique non-trivial self-isometry of O^3 .*

Proof. Applying Lemma 2.4 to the permutation representation

$$\rho_f \mapsto [0, 2, 1], \quad \rho_e \mapsto [1, 0, 2], \quad \rho_v \mapsto [0, 2, 1],$$

we take $T_3 \cup \rho_e(T_3) \cup \rho_e\rho_v(T_3)$ as a fundamental domain for H^3 . Its intersection with the plane $\mathcal{H}$ over $\mathbb{R}$ is pictured on the left in Figure 3. Of the open faces of


 FIGURE 3. Intersections with the plane $\mathcal{H}$ over $\mathbb{R}$, for O^3 (left) and $\tilde{Q}_{\sqrt{-3}}$.

this fundamental domain, $\rho_f \in H^3$ pairs f_- to f_+ ; ρ_v pairs $v \cup \rho_e \rho_v(e)$ to itself; and $\rho_e \rho_f \rho_v \rho_e$ and $\rho_e \rho_v \rho_f \rho_e$ respectively pair $\rho_e \rho_v(f_-)$ to $\rho_e(f_+)$ and $\rho_e(f_-)$ to $\rho_e \rho_v(f_+)$. The two ideal vertices ∞ and 1 of $T_3 \cup \rho_e(T_3) \cup \rho_e \rho_v(T_3)$ are identified under these face-pairings, and the resulting cusp of O^3 has an $S^2 : (2, 3, 6)$ cross-section, in the taxonomy of [19, Th. 13.3.6].

The given fundamental domain is symmetric under the reflection r , and as r conjugates ρ_f to ρ_f^{-1} and ρ_e and ρ_v to themselves, it normalizes H^3 . Specifically, it conjugates $\rho_e \rho_f \rho_v \rho_e$ to $\rho_e \rho_f^{-1} \rho_v \rho_e = \rho_e \rho_v \rho_f \rho_e$, since $(\rho_f \rho_v)^2 = 1$, and vice-versa. This is the only non-trivial isometry of O^3 since $S^2 : (2, 3, 6)$ already has no other isometries. $\square$

2.3. The second-smallest minimal orbifold in $\mathcal{C}_3$. This subsection concerns the minimal orientable orbifold in $\mathcal{C}_3$, termed $Q_{\sqrt{-3}}$ here and Q_3 in [16], with volume $v_0/6$. As described in [16], $Q_{\sqrt{-3}} = \mathbb{H}^3/\Gamma_{\sqrt{-3}}$ for a group $\Gamma_{\sqrt{-3}}$ having an index-two subgroup $\tilde{\Gamma}_{\sqrt{-3}}$, its intersection with $\text{PGL}_2(O_3)$, that is a tetrahedral rotation group. The results of the present subsection give presentations and more data for $\Gamma_{\sqrt{-3}}$ and $\tilde{\Gamma}_{\sqrt{-3}}$. We summarize them as follows.

Lemma 2.7. *For $H^4 < \text{PGL}_2(O_3)$ and the degree-four cover $O^4 = \mathbb{H}^3/H^4$ of $Q_0 = \mathbb{H}^3/\text{PGL}_2(O_3)$ as in Definition 2.3, O^4 is tetrahedral. Specifically, H^4 is isomorphic to the rotation group in $T(2, 2, 3; 2, 6, 3)$. Re-naming H^4 as $\tilde{\Gamma}_{\sqrt{-3}}$ and taking $\alpha = \rho_f$, $\beta = \rho_e$, and $\gamma = \rho_v \rho_e \rho_f^2 \rho_e \rho_v$, for ρ_f , ρ_e , and ρ_v as in (2), $\tilde{\Gamma}_{\sqrt{-3}} = \langle \alpha, \beta, \gamma \rangle$ has presentation:*

$$\tilde{\Gamma}_{\sqrt{-3}} \cong \langle \alpha, \beta, \gamma \mid \alpha^6 = \beta^2 = \gamma^3 = (\alpha\beta)^3 = (\alpha\gamma^{-1})^2 = (\beta\gamma^{-1})^2 = 1 \rangle.$$

The reflection group in $T(2, 2, 3; 2, 6, 3)$ is generated by $\tilde{\Gamma}_{\sqrt{-3}}$ and the reflection r in the plane $\mathcal{H}$ over $\mathbb{R}$.

There is an extension $\Gamma_{\sqrt{-3}} \cong \tilde{\Gamma}_{\sqrt{-3}} \rtimes \mathbb{Z}/2\mathbb{Z}$ of $\tilde{\Gamma}_{\sqrt{-3}}$ that is not contained in $\text{PGL}_2(O_3)$, presented as:

$$\Gamma_{\sqrt{-3}} \cong \langle \alpha, \beta, \gamma, \mu \mid \alpha^6 = \beta^2 = \gamma^3 = \mu^2 = (\alpha\beta)^3 = (\alpha\gamma^{-1})^2 = (\beta\gamma^{-1})^2 = 1, \\ \mu\alpha\mu = \alpha^{-1}, \mu\beta\mu = \alpha\gamma^{-1}, \mu\gamma\mu = (\alpha\beta)^{-1} \rangle$$

Here μ is an order-two rotation of $T(2, 2, 3; 2, 6, 3)$ (notation as in [16]) which also normalizes the group generated by reflections in its faces. Together, μ and this reflection group generate a group $\tilde{\Gamma}_{\sqrt{-3}}$ that contains $\Gamma_{\sqrt{-3}}$ with index two.

Remark. The notations $\tilde{\Gamma}_{\sqrt{-3}}$ and $\Gamma_{\sqrt{-3}}$, for groups, and $\tilde{Q}_{\sqrt{-3}}$ and $Q_{\sqrt{-3}}$, for orbifolds, place these in the broader context of minimal orbifolds in the commensurability class described in Proposition 3.1 and Theorem 3.4.

Recall that $\tilde{\Gamma}_{\sqrt{-3}} \doteq H^4$ is the stabilizer of 0 under the right-permutation representation $\mathrm{PGL}_2(O_3) \rightarrow S_4$ determined by

$$\rho_f \mapsto [0, 1, 3, 2], \quad \rho_e \mapsto [0, 2, 1, 3], \quad \rho_v \mapsto [1, 0, 2, 3].$$

We now use Lemma 2.4 to recover the tetrahedron $T(2, 2, 3; 2, 6, 3)$ and a generating set for $\tilde{\Gamma}_{\sqrt{-3}}$ in terms of the generators for $\mathrm{PGL}_2(O_3)$. From the permutation representation we obtain the following right actions: $0.\rho_v = 1$, $0.(\rho_v\rho_e) = 2$, $0.(\rho_v\rho_e\rho_f) = 3$. Therefore the union of the translates T_3 , $\rho_v(T_3)$, $\rho_v\rho_e(T_3)$, $\rho_v\rho_e\rho_f(T_3)$ is a fundamental domain for the action of $\tilde{\Gamma}_{\sqrt{-3}}$.

Because each of ρ_v and ρ_e preserves the plane $\mathcal{H}$ in $\mathbb{H}^3$ above $\mathbb{R}$, each of $\rho_v(T_3)$ and $\rho_v\rho_e(T_3)$ shares the property with T_3 that this plane bisects it into sub-tetrahedra. We picture these cells' intersections with $\mathcal{H}$ in Figure 3. Because ρ_f rotates counterclockwise around the edge it fixes—when orienting this edge pointing toward its ideal vertex—the final tile $\rho_v\rho_e\rho_f(T_3)$ lies on the “positive” side of $\mathcal{H}$ —consisting of those points in $\mathbb{H}^3$ whose complex coordinates have positive imaginary part. It intersects $\rho_v\rho_e(T_3)$ in that cell's face $\rho_v\rho_e(f_+)$ and its own face $\rho_v\rho_e\rho_f(f_-)$.

The tetrahedron $T(2, 2, 3; 2, 6, 3)$ is the intersection of these four translates' union with a cell of the tessellation $\mathcal{T}_3$ —the regular ideal tetrahedron with vertices at 0, 1, ∞ and $\frac{1}{2}(1 + i\sqrt{3})$. We can create a symmetric fundamental domain for $\tilde{\Gamma}_{\sqrt{-3}}$, consisting of the union of $T(2, 2, 3; 2, 6, 3)$ and its reflection across $\mathcal{H}$, by cutting $\rho_v\rho_e\rho_f(T_3)$ in half along $\rho_v\rho_e\rho_f(T_3 \cap \mathcal{H})$ and moving the half outside $T(2, 2, 3; 2, 6, 3)$ to the negative side via the conjugate of ρ_f^{-2} by $\rho_v\rho_e$, ie. $\rho_v\rho_e\rho_f^{-2}\rho_e\rho_v$. With this fundamental domain, we have the set of face pairing generators:

$$\tilde{\Gamma}_{\sqrt{-3}} = \langle \rho_f, \rho_e, \rho_v\rho_e\rho_f^2\rho_e\rho_v \rangle.$$

These elements rotate around the three bold sides of the fundamental domain's intersection with $\mathcal{H}$, as seen in Figure 3, in each case taking a face of the symmetrized fundamental domain on the negative side to its mirror image on the positive side, thus generating the tetrahedral rotation group. Since the reflection r is in the plane $\mathcal{H}$ containing a face of $T(2, 2, 3; 2, 6, 3)$, it and $\tilde{\Gamma}_{\sqrt{-3}}$ together generate the tetrahedral reflection group.

Taking $\alpha \doteq \rho_f$, $\beta \doteq \rho_e$, and $\gamma \doteq \rho_v\rho_e\rho_f^2\rho_e\rho_v$, we obtain the presentation of Lemma 2.7 for $\tilde{\Gamma}_{\sqrt{-3}}$ from the Poincaré polyhedron theorem. This uses the fact that the edges of $T(2, 2, 3; 2, 6, 3)$ not pictured in Figure 3 have dihedral angles $\pi/3$ (with an endpoint at ∞), $\pi/2$, and $\pi/2$ (the other two), each belonging to a cycle of two edges.

As observed in [16], $T(2, 2, 3; 2, 6, 3)$ has a rotational involution μ that preserves its edge with dihedral angle $\pi/6$ (joining 0 to ∞ here) and the opposite edge—a compact one with dihedral angle $\pi/2$ fixed by $\gamma\beta$. Thus μ conjugates each of $\alpha = \rho_f$ and $\gamma\beta$ to its inverse, and conjugation by μ exchanges the elliptic elements fixing the other four edges of $T(2, 2, 3; 2, 6, 3)$ in pairs: it takes β to $\alpha\gamma^{-1}$ and γ to $(\alpha\beta)^{-1}$. (The reason μ conjugates γ to the *inverse* of $\alpha\beta$ is that γ rotates counterclockwise around its fixed edge, oriented from its compact to its ideal vertex, whereas $\alpha\beta$ rotates clockwise around its fixed edge with the corresponding orientation.)

Thus describing $\Gamma_{\sqrt{-3}} = \langle \tilde{\Gamma}, \mu \rangle$ as a semidirect product we obtain the presentation of Lemma 2.7. Because μ preserves $T(2, 2, 3; 2, 6, 3)$ it also normalizes the group generated by reflections in its faces, generating a group $\bar{\Gamma}_{\sqrt{-3}}$ together with them which contains $\Gamma_{\sqrt{-3}}$ with index two.

We note that μ does not belong to $\mathrm{PGL}_2(O_3)$ since it does not preserve the $\mathrm{PGL}_2(O_3)$ -invariant tiling by copies of T_3 . We can see this from $T(2, 2, 3; 2, 6, 3)$, since μ preserves it and exchanges its vertices at ∞ —belonging to only T_3 itself among its translates in $T(2, 2, 3; 2, 6, 3)$ —with 0, which belongs to three translates. Therefore $\Gamma_{\sqrt{-3}}$ is not contained in $\mathrm{PGL}_2(O_3)$.

2.4. The third-smallest minimal orbifold in $\mathcal{C}_3$. This subsection concerns the minimal orientable orbifold in $\mathcal{C}_3$, termed Q_2 here and Q_4 in [16], with volume $5v_0/12$. As described in [16], $Q_2 = \mathbb{H}^3/\Gamma_2$ for a tetrahedral rotation group Γ_2 generated by rotations in the edges of the tetrahedron $T(4, 2, 2, 6, 2, 3)$. Also from [16], the subgroup $\tilde{\Gamma}_2 = \Gamma_2 \cap \mathrm{PGL}_2(O_3)$ has index two in Γ_2 and index five in $\mathrm{PGL}_2(O_3)$. The results of the present subsection give presentations and more data for Γ_2 and $\tilde{\Gamma}_2$.

Lemma 2.8. *For $H^5 < \mathrm{PGL}_2(O_3)$ and the degree-five cover $O^5 = \mathbb{H}^3/H^5$ of $Q_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_3)$ as in Definition 2.3, O^5 is tetrahedral. Specifically, H^5 is isomorphic to the rotation group in $T(2, 3, 2; 6, 2, 3)$. Re-naming H^5 as $\tilde{\Gamma}_2$, and taking $\alpha_2 = \rho_f$, $\beta_2 = \rho_e$, and $\gamma_2 = \rho_v \rho_e \rho_f^3 \rho_e \rho_v$, for ρ_f , ρ_e , and ρ_v as in (2), $\tilde{\Gamma}_2 = \langle \alpha_2, \beta_2, \gamma_2 \rangle$ has presentation:*

$$\tilde{\Gamma}_2 \cong \langle \alpha_2, \beta_2, \gamma_2 \mid \alpha_2^6 = \beta_2^2 = \gamma_2^2 = (\alpha_2 \beta_2)^3 = (\alpha_2 \gamma_2)^3 = (\beta_2 \gamma_2)^2 = 1 \rangle.$$

The reflection group in $T(2, 3, 2; 6, 2, 3)$ is generated by $\tilde{\Gamma}_2$ and the reflection r in the plane $\mathcal{H}$ over $\mathbb{R}$.

The tetrahedral rotation group $T(4, 2, 2; 6, 2, 3)$ is isomorphic to an extension $\Gamma_2 \cong \tilde{\Gamma}_2 \rtimes \mathbb{Z}/2\mathbb{Z}$ of $\tilde{\Gamma}_2$ that does not lie in $\mathrm{PGL}_2(O_3)$. It is presented as:

$$\Gamma_2 \cong \langle \alpha_2, \gamma_2, \mu_2 \mid \alpha_2^6 = \gamma_2^2 = \mu_2^2 = (\alpha_2 \gamma_2)^3 = (\alpha_2 \mu_2)^2 = (\mu_2 \gamma_2)^4 = 1 \rangle$$

Here μ_2 is the order-two rotation around an axis in $\mathcal{H}$ that meets the geodesic from 0 to ∞ at right angles and runs through the intersection of the axes of β_2 and γ_2 . Conjugation by μ_2 exchanges β_2 with γ_2 .

Remark. The notations $\tilde{\Gamma}_2$ and Γ_2 , for groups, and $\tilde{Q}_2$ and Q_2 , for orbifolds, place these in the broader context of minimal orbifolds in the commensurability class described in Proposition 3.1 and Theorem 3.4.

Recall that $\tilde{\Gamma}_2 \doteq H^5$ is the stabilizer of 0 under the permutation representation $\mathrm{PGL}_2(O_3) \rightarrow S_5$ determined by

$$\rho_f \mapsto [0, 1, 3, 4, 2], \quad \rho_e \mapsto [0, 2, 1, 4, 3], \quad \rho_v \mapsto [1, 0, 2, 4, 3]$$

The following elements give a full set of right coset representatives for the stabilizer $\tilde{\Gamma}_2$ of 0, together with the identity:

$$0. \rho_v = 1, \quad 0. (\rho_v \rho_e) = 2, \quad 0. (\rho_v \rho_e \rho_f) = 3, \quad 0. (\rho_v \rho_e \rho_f^{-1}) = 4.$$

By Lemma 2.4, the union of T_3 with its translates under ρ_v , $\rho_v \rho_e$, $\rho_v \rho_e \rho_f$, and $\rho_v \rho_e \rho_f^{-1}$ thus constitutes a fundamental domain for the action of $\tilde{\Gamma}_2$ on $\mathbb{H}^3$. All of these but the last form the fundamental domain for $\tilde{\Gamma}_{\sqrt{-3}}$ discussed in the previous

subsection, and we refer to the exposition there (and Figure 3) for their locations. Building on that, we observe that the final translate $\rho_v \rho_e \rho_f^{-1}(T_3)$ intersects $\rho_v \rho_e(T_3)$ along its face $\rho_v \rho_e(f_-)$ and is symmetric to $\rho_v \rho_e \rho_f(T_3)$ across $\mathcal{H}$.

Since each of T_3 , $\rho_v(T_3)$, and $\rho_v \rho_e(T_3)$ is symmetric across $\mathcal{H}$, as was previously observed, their union with $\rho_v \rho_e \rho_f^{-1}(T_3)$ is as well. The half of this union on the positive side of $\mathcal{H}$ is a tetrahedron. Its edges with endpoints at ∞ contain those of T_3^+ and thus have dihedral angles $\pi/6$, $\pi/2$, and $\pi/3$. The face opposite this ideal vertex is the face $\rho_v \rho_e \rho_f(f_+)$ of $\rho_v \rho_e \rho_f(T_3)$. The edges of this face that do not lie in $\mathcal{H}$ are not shared by $\rho_v \rho_e \rho_f(T_3)$ with any other translate in the fundamental domain, so their dihedral angles are $\pi/3$ (the edge containing the ideal vertex 0) and $\pi/2$ (the compact edge). For the edge contained in $\mathcal{H}$, which is shared in the fundamental domain for $\tilde{\Gamma}_3$ by $\rho_v \rho_e \rho_f^{-1}(T_3)$ and $\rho_v \rho_e(T_3)$, its dihedral angle in the “positive half” of this fundamental domain is thus $\pi/2$.

The tetrahedron described in the paragraph above is thus $T(2, 3, 2; 6, 2, 3)$ in the notation of [16], and applying the Poincaré polyhedron theorem to the face-pairing generators for the entire fundamental domain which rotate around its edges in $\mathcal{H}$ yields the presentation of Lemma 2.8. Since the reflection r is in the plane $\mathcal{H}$ containing a face of $T(2, 3, 2; 6, 2, 3)$, it and $\tilde{\Gamma}_2$ together generate the tetrahedral reflection group.

There is a combinatorial reflective symmetry of $T(2, 3, 2; 6, 2, 3)$ that interchanges the endpoints of the edge labeled 6 and fixes its opposite edge (labeled 2). By a standard application of Andreev’s theorem, this implies that $T(2, 3, 2; 6, 2, 3)$ is divided into mirror-image copies of the tetrahedron $T(4, 2, 2; 6, 2, 3)$ mentioned in [16], which meet along the face with dihedral angles $\pi/4$, $\pi/2$, and $\pi/2$.

With $T(2, 3, 2; 6, 2, 3)$ embedded as described above, as the “positive half” of the fundamental domain

$$T_3 \cup \rho_v(T_3) \cup \rho_v \rho_e(T_3) \cup \rho_v \rho_e \rho_f(T_3) \cup \rho_v \rho_e \rho_f^{-1}(T_3)$$

for $\tilde{\Gamma}_2$, its intersection with $\mathcal{H}$ is the union of the three bold triangles in Figure 3. (I.e. the same as for the tetrahedron $T(2, 2, 3; 2, 6, 3)$ considered in the previous subsection.) The two copies of $T(4, 2, 2; 6, 2, 3)$ in $T(2, 3, 2; 6, 2, 3)$ each intersect $\mathcal{H}$ in half of this face, divided by a geodesic arc that bisects the angle at its finite vertex and intersects the opposite, bi-infinite side, perpendicularly. The dihedral angle of $T(4, 2, 2; 6, 2, 3)$ around this edge is thus $\pi/2$.

Let μ_2 be the order-two rotation about the edge of $T(4, 2, 2; 6, 2, 3)$ described above, which is properly embedded in the face of $T(2, 3, 2; 6, 2, 3)$ contained in $\mathcal{H}$. The rotation group in $T(4, 2, 2; 6, 2, 3)$ is generated by μ_2 and the rotations around its other two edges contained in $\mathcal{H}$. If we take the copy of $T(4, 2, 2; 6, 2, 3)$ having its ideal vertex at 0, these rotations are α_2 and γ_2 . The presentation given in Lemma 2.8 follows in a standard way by considering the dihedral angles of $T(4, 2, 2; 6, 2, 3)$.

We can already see that μ_2 is not contained in $\text{PGL}_2(O_3)$ by the fact that it takes the face of $T(2, 3, 2; 6, 2, 3)$ pictured in Figure 3 to itself but does not preserve its triangulation pictured in the Figure—due eg. to the fact that μ_2 exchanges 0 and ∞ , which have different valences in this triangulation. But this triangulation is contained in the $\text{PGL}_2(O_3)$ -invariant tessellation consisting of $\text{PGL}_2(O_3)$ -translates of its fundamental domain.

2.5. Index-two subgroups and reflections. The main result of this subsection describes the index-two subgroups $\Gamma_{\sqrt{-3}}$ and Γ_2 , and the action on them of the reflection r of $\mathbb{H}^3$ that extends complex conjugation on $\mathbb{C}$.

Lemma 2.9. *Each of $\Gamma_{\sqrt{-3}}$ and Γ_2 has three subgroups of index two, corresponding to kernels of the homomorphisms to $\mathbb{Z}/2\mathbb{Z} = \{\pm 1\}$ given below:*

For $\Gamma_{\sqrt{-3}}$ $\phi_0 : \alpha, \beta, \gamma \mapsto 1, \mu \mapsto -1$ $\phi_1 : \gamma, \mu \mapsto 1, \alpha, \beta \mapsto -1$ $\phi_2 : \gamma \mapsto 1, \alpha, \beta, \mu \mapsto -1$	For Γ_2 $\psi_0 : \alpha_2, \gamma_2 \mapsto 1, \mu_2 \mapsto -1$ $\psi_1 : \alpha_2, \gamma_2 \mapsto -1, \mu_2 \mapsto 1$ $\psi_2 : \alpha_2, \gamma_2, \mu_2 \mapsto -1$
---	--

(5)

The kernels of ϕ_0 and ψ_0 are respectively equal to $\tilde{\Gamma}_{\sqrt{-3}}$ from Lemma 2.7 and $\tilde{\Gamma}_2$ from Lemma 2.8.

All three $\ker \psi_i$ are normalized by the reflection r of $\mathbb{H}^3$ fixing the plane $\mathcal{H}$ over $\mathbb{R}$, as is $\tilde{\Gamma}_{\sqrt{-3}}$. The other two index-two subgroups of $\Gamma_{\sqrt{-3}}$ are conjugate by r .

The Lemma's lists are obtained using the presentations of Lemmas 2.7 and 2.8. By the well known fact that all index-two subgroups correspond to homomorphisms onto $\mathbb{Z}/2\mathbb{Z}$, these give all index-two subgroups of each group.

The kernels of ϕ_0 and ψ_0 are $\tilde{\Gamma}_{\sqrt{-3}}$ and $\tilde{\Gamma}_2$, respectively, since in each case the generators map trivially. Because each generator for either of $\tilde{\Gamma}_{\sqrt{-3}}$ and $\tilde{\Gamma}_2$ rotates around an axis in $\mathcal{H}$, the reflection r in $\mathcal{H}$ conjugates it to its inverse. It follows that r normalizes each of $\tilde{\Gamma}_{\sqrt{-3}}$ and $\tilde{\Gamma}_2$.

Because the axis of μ_2 also lies in $\mathcal{H}$, r also conjugates μ_2 to its inverse (itself). The fact that r normalizes $\ker \psi_1$ and $\ker \psi_2$ will follow from a description of their generating sets. Using α_2 as the non-trivial coset representative for each in Γ_2 , the Schreier Lemma gives the following generating sets:

$$\ker \psi_1 = \langle \alpha_2^2, \gamma_2 \alpha_2^{-1}, \alpha_2 \gamma_2, \mu_2, \alpha_2 \mu_2 \alpha_2^{-1} \rangle, \ker \psi_2 = \langle \alpha_2^2, \gamma_2 \alpha_2^{-1}, \alpha_2 \gamma_2, \mu_2 \alpha_2^{-1}, \alpha_2 \mu_2 \rangle$$

We thus easily find that each kernel above is normalized by r , via computations like this: $r(\gamma_2 \alpha_2^{-1})r = \gamma_2^{-1} \alpha_2 = (\alpha_2 \gamma_2)^{-1} \alpha_2^2$.

The story for $\ker \phi_1$ and $\ker \phi_2$ is different. As observed above, r conjugates each of α , β , and γ to its inverse; however we claim that r conjugates the generator μ of $\Gamma_{\sqrt{-3}}$ to $\mu\alpha$. Together with the following generating sets for $\ker \phi_1$ and $\ker \phi_2$ (which use α as the non-trivial coset representative for each), this will imply that r conjugates $\ker \phi_1$ to $\ker \phi_2$.

$$\begin{aligned} \ker \phi_1 &= \langle \gamma, \mu, \alpha^2, \alpha\beta, \beta\alpha^{-1}, \alpha\gamma\alpha^{-1}, \alpha\mu\alpha^{-1} \rangle \\ \ker \phi_2 &= \langle \gamma, \alpha^2, \alpha\beta, \alpha\mu, \beta\alpha^{-1}, \mu\alpha^{-1}, \alpha\gamma\alpha^{-1} \rangle \end{aligned}$$

To prove the claim, we note that the fixed axis of μ intersects the edge of $T(2, 2, 3; 2, 6, 3)$ joining 0 to ∞ perpendicularly, bisecting the angle made by the faces meeting along this edge. This axis thus has an angle of $\pi/12$ from the face of $T(2, 2, 3; 2, 6, 3)$ in $\mathcal{H}$. If we restrict to the plane Π that is perpendicular to the geodesic between 0 and ∞ and contains the fixed axis of μ , the restriction of μ acts by reflection across its axis, and α rotates counterclockwise (viewed from ∞) by an angle of $\pi/3$ around the point o of intersection between this axis and the geodesic from 0 to ∞ . Thus, μ and α generate a dihedral group fixing o , and $\mu\alpha$ fixes an axis in Π at a clockwise angle of $\pi/6$ around o from the axis of μ —its image under

r , which fixes the axis $\mathcal{H} \cap \Pi$ at a clockwise angle of $\pi/12$ from that of μ . This implies the claim, and hence establishes Lemma 2.9.

Lemma 2.10. *For ϕ_i and ψ_i as in (5), $i = 1, 2$, let $P_{\sqrt{-3}}^i = \mathbb{H}^3 / \ker \phi_i$ and $P_2^i = \mathbb{H}^3 / \ker \psi_i$. For $i = 1, 2$, $P_{\sqrt{-3}}^i$ has no orientation-reversing self-isometry.*

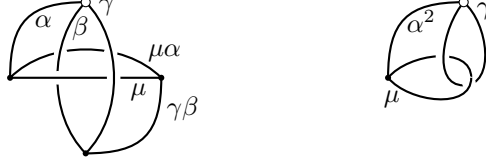


FIGURE 4. The singular loci of $Q_{\sqrt{-3}}$ (left) and $P_{\sqrt{-3}}^1$ (right).

Proof. The singular locus of $Q_{\sqrt{-3}}$ is pictured on the left in Figure 4, with each edge labeled by a generator of the corresponding conjugacy class of cyclic elliptic subgroups of $\Gamma_{\sqrt{-3}}$. This is produced by taking the quotient of $\tilde{Q}_{\sqrt{-3}}$, which we recall is tetrahedral, by the self-isometry induced by μ .

For a cyclic elliptic subgroup G contained in the index-two subgroup $\ker \phi_1$ —here, $G = \langle \gamma \rangle$ or $\langle \mu \rangle$ —the conjugacy class of G in $Q_{\sqrt{-3}}$ divides into at most two conjugacy classes in $\ker \phi_1$, with the second being that of gGg^{-1} for some (any) $g \notin \ker \phi_1$, if this is not conjugate in $\ker \phi_1$ to G . In the present case, since $\beta\gamma\beta^{-1} = \gamma^{-1}$ and μ commutes with $\gamma\beta$, the conjugacy classes of each of $\langle \gamma \rangle$ and $\langle \mu \rangle$ are identical in $\Gamma_{\sqrt{-3}}$ and $\ker \phi_1$.

For a cyclic elliptic subgroup $G \not\subset \ker \phi_1$ of order greater than two, the square of its generator g determines a unique conjugacy class of cyclic elliptic subgroups of $\ker \phi_1$. This is because for any $h \notin \ker \phi_1$, $hg^2h^{-1} = (hg)g^2(g^{-1}h^{-1})$ is the conjugate of g^2 by the element hg of $\ker \phi_1$. Here, $\langle \alpha^2 \rangle$ determines such a conjugacy class of order-three subgroups of $\ker \phi_1$. The cyclic elliptic subgroups of order two that are not contained in $\ker \phi_1$ —generated by β , $\mu\alpha$, and $\gamma\beta$ —do not determine elliptic subgroups of $\ker \phi_1$. Combining this information, we obtain the singular locus for $P_{\sqrt{-3}}^1$ pictured on the right in Figure 4.

Any self-isometry of $P_{\sqrt{-3}}^1$ preserves its cusp and singular locus. In particular, it must take the half-ideal edge corresponding to α^2 to itself. As a result, for an element g of the normalizer of $\ker \phi_1$ in the isometry group of $\mathbb{H}^3$, after replacing g by gh for some $h \in \ker \phi_1$, we may assume that g fixes each of ∞ and the origin. If g reverses orientation, it is then a reflection in a vertical plane containing the origin. This reflection must carry the axis of μ to another elliptic axis sharing its fixed point on the geodesic from 0 to ∞ . There are three of these, evenly spaced at angles of $\pi/3$: those of μ , $\alpha^2\mu$, and $\alpha^{-2}\mu$.

Such a reflection must also preserve the set of fixed points of elliptic elements that also fix ∞ . The stabilizer of ∞ in $\ker \phi_1$ is generated by α^2 and $\alpha\beta$, the latter an order-three rotation fixing $\frac{1}{2} \left(1 + i\frac{\sqrt{3}}{3}\right)$, conjugate by μ to γ . The other five elliptic fixed points closest to 0 are spaced at angles of $\pi/3$, each from the next, around a circle of radius $\frac{1}{\sqrt{3}}$ centered at 0. The angle separation between the sets of elliptic fixed points and elliptic axes is not compatible with the existence of a single

reflection that simultaneously preserves both: from Lemma 2.9, the axis of μ is at an angle of $\pi/12$ from the plane $\mathcal{H}$ over $\mathbb{R}$, whereas the fixed point of $\alpha\beta$ is at an angle of $\pi/6$ from this plane. Hence there is no orientation-reversing self-isometry of $P_{\sqrt{-3}}^1$. $\square$

We close with further information about the P_2^i .

Lemma 2.11. *The subgroup $\ker \psi_2$ of Γ_2 is isomorphic to the tetrahedral rotation group of $T(4, 3, 3; 3, 2, 2)$; hence P_2^2 is tetrahedral. The orbifold P_2^1 is not tetrahedral.*

Proof. Recall from Lemma 2.8 that Γ_2 and $\tilde{\Gamma}_2$ are the tetrahedral rotation groups of $T(4, 2, 2; 6, 2, 3)$ and $T(2, 3, 2; 6, 2, 3)$, respectively. The latter tetrahedron is the double of the former across its face with edges having dihedral angles $\pi/4$, $\pi/2$, and $\pi/2$. A second face of $T(4, 2, 2; 6, 2, 3)$ has two edges that have dihedral angles of $\pi/2$, and an angle of $\pi/6$ along its third edge. With $T(4, 2, 2; 6, 2, 3)$ arranged as in the proof of Lemma 2.8, this face lies in $\mathcal{H}$. Doubling $T(4, 2, 2; 6, 2, 3)$ across it thus yields the tetrahedron $T(4, 3, 3; 3, 2, 2)$ mentioned in the Lemma; its tetrahedral rotation group is therefore an index-two subgroup of that of $T(4, 2, 2; 6, 2, 3)$, Γ_2 . Because the generators α_2 , γ_2 , μ_2 of Γ_2 belong to the face that is doubled across to produce $T(4, 3, 3; 3, 2, 2)$, they do not belong to its tetrahedral rotation group. This subgroup is therefore $\ker \psi_2$.

Refer to $T(4, 3, 3; 3, 2, 2)$ as T for short. This tetrahedron is a fundamental domain for Γ_2 , with μ_2 and γ_2 each pairing a face to itself, rotating by π around a bisector of that face. The final generator α_2 pairs the remaining two faces of T , rotating about their edge of intersection. Applying Lemma 2 to the right-permutation representation of Γ_2 corresponding to ψ_1 , which sends α_2 and γ_2 to $[1, 0]$ and μ_2 to $[0, 1]$, we take $T \cup \gamma_2(T)$ as a fundamental domain. The Lemma gives μ_2 , $\gamma_2\mu_2\gamma_2^{-1}$, $\alpha_2\gamma_2$, and $\gamma_2\alpha_2$ as face-pairings for this domain. From this we discern the singular locus of P_2^1 , pictured in Figure 5. As its singular locus is not a complete graph, P_2^1 is not tetrahedral. $\square$

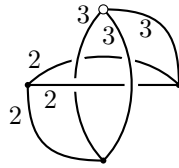


FIGURE 5. The singular locus of P_2^1

3. MINIMAL ORBIFOLDS IN THE COMMENSURABILITY CLASS

We call an (orientable) orbifold *minimal* if it does not cover any other (orientable) orbifold. It follows from work of Margulis that any non-arithmetic commensurability class contains a unique minimal orbifold; in contrast, Borel [3] gave a general classification of the infinitely many minimal orientable 3-orbifolds in an arithmetic commensurability class. Here we will consider non-compact arithmetic hyperbolic 3-orbifolds, each of which is commensurable to a “Bianchi orbifold” $\mathbb{H}^3/\mathrm{PSL}_2(O_d)$ for some positive square-free integer d , where O_d is the ring of integers of $\mathbb{Q}(\sqrt{-d})$.

Specializing further to the case that $\mathbb{Q}(\sqrt{-d})$ has class number one (which encompasses the sub-cases $d = 1$ and $d = 3$ of most interest here), we use the account of Borel's work given in [13], and the subsequent work [5], to explicitly describe the minimal orbifolds in the commensurability class of $\mathrm{PSL}_2(O_d)$, their smallest common covers with $\mathbb{H}^3/\mathrm{PGL}_2(O_d)$, and these orbifolds' cusps. The class number one condition is equivalent to O_d being a principal ideal domain; in this case the minimal orientable orbifolds commensurable but not equal to $\mathbb{H}^3/\mathrm{PGL}_2(O_d)$ correspond to products π of distinct primes of O_d . For any such π , we call the associated minimal orientable orbifold Q_π , and we call $\tilde{Q}_\pi$ its smallest common cover with $\mathbb{H}^3/\mathrm{PGL}_2(O_d)$. For $\mathbb{Q}(\sqrt{-d})$ having class number one, the main result of [5] implies in particular that each Q_π above has one cusp.

Section 3.1 gives a reasonably complete and self-contained account of the work above applied to our case, culminating in Proposition 3.1. Section 3.3 describes the cusps of the Q_π and $\tilde{Q}_\pi$, and the covering maps between them. In Section 3.2, further specializing to the case $d = 3$, we establish Theorem 3.4, which gives the classification of minimal orbifolds in $\mathcal{C}_3$ that have volume at most $2v_0$, by appealing to results of the prior subsections and computing the norms of small primes in O_3 .

3.1. The Classification of Minimal Orbifolds. Here we will describe the machinery for enumerating maximal elements of a commensurability class determined by a maximal order $\mathcal{O}$ in a quaternion algebra A over a number field k , restricting to the case that $k = \mathbb{Q}(\sqrt{-d})$ for a square-free positive integer d ; $A = M_2(\mathbb{Q}(\sqrt{-d}))$ is the algebra of 2×2 matrices with entries in k ; and $\mathcal{O} = M_2(O_d)$, where O_d is the ring of integers of k . For general background on orders in quaternion algebras see eg. Maclachlan–Reid [13, Ch. 2].

Below we sync notation with [13, §11.4], which presents the results of [3] with expanded background and exposition. As described there, each maximal group in the commensurability class is conjugate to some $\Gamma_{S,\mathcal{O}} = P(N(\mathcal{E}))$ from [13, Dfn. 11.4.2]. Here $\mathcal{E} = \mathcal{O} \cap \mathcal{O}'$ is an *Eichler order*, the intersection of $\mathcal{O}$ and another maximal order $\mathcal{O}' = \mathcal{O}'(S)$ of A , where S is a finite set of prime ideals of O_d ; $N(\mathcal{E})$ is the normalizer of $\mathcal{E}$, consisting of the invertible elements of A that conjugate $\mathcal{E}$ to itself; and P is the projectivization.

The maximal order $\mathcal{O}'$ determined by S is characterized by the property that its localization $\mathcal{O}'_{\mathcal{P}}$ at any prime ideal $\mathcal{P} \notin S$ coincides with that of $\mathcal{O}$, and for any $\mathcal{P} \in S$, $\mathcal{O}'_{\mathcal{P}}$ stabilizes a vertex at distance one from that of $\mathcal{O}_{\mathcal{P}}$ in the tree for $\mathrm{SL}_2(k_{\mathcal{P}})$, cf. [13, Lemma 6.2.7]. Here $k_{\mathcal{P}}$ is the completion of k with respect to the metric induced by the $\mathcal{P}$ -adic valuation $v_{\mathcal{P}}$ on k , which extends continuously to $k_{\mathcal{P}}$; $R_{\mathcal{P}} = \{x \in k_{\mathcal{P}} \mid v_{\mathcal{P}}(x) \leq 1\}$ is the *ring of $\mathcal{P}$ -adic integers* (see eg. [13, §0.7]); and the algebra $A = M_2(k)$ and order $\mathcal{O} \subset A$ are *localized at $\mathcal{P}$* by $A_{\mathcal{P}} = k_{\mathcal{P}} \otimes_k A$ and $\mathcal{O}_{\mathcal{P}} = R_{\mathcal{P}} \otimes_{O_d} \mathcal{O}$ [13, Dfn. 6.2.6]. We describe $\mathcal{O}'$ more explicitly below after recalling some background on the tree $\mathcal{T}_{\mathcal{P}}$ for $\mathrm{SL}_2(k_{\mathcal{P}})$, cf. [13, p. 211].

For any prime ideal $\mathcal{P}$, the vertices of $\mathcal{T}_{\mathcal{P}}$ are equivalence classes under scaling of complete $R_{\mathcal{P}}$ -lattices in a fixed two-dimensional vector space V over $k_{\mathcal{P}}$. An edge joins any two vertices represented by lattices $L' \subset L$ such that L/L' is isomorphic to the residue field $R_{\mathcal{P}}/\pi R_{\mathcal{P}}$, where π is a *uniformizer*, generating the unique maximal ideal of $R_{\mathcal{P}}$. In the case that $R = O_d$ is a PID, any generator $\pi \in O_d$ of $\mathcal{P}$ is a uniformizer of $R_{\mathcal{P}}$; and every element of $O_d - \mathcal{P}$ is a unit in $R_{\mathcal{P}}$.

As $A_{\mathcal{P}} \cong M_2(k_{\mathcal{P}})$ acts on V , its group of invertible elements $\mathrm{GL}_2(k_{\mathcal{P}})$ acts on $\mathcal{T}_{\mathcal{P}}$. As scalar matrices act trivially, this action factors through the projectivization.

For each maximal order $\mathcal{O}_{\mathcal{P}} \subset A_{\mathcal{P}}$, its group of units $\mathcal{O}_{\mathcal{P}}^1$ is a vertex stabilizer for $\mathcal{T}_{\mathcal{P}}$; the resulting correspondence with vertices of $\mathcal{T}_{\mathcal{P}}$ is bijective [13, Lemma 6.5.1].

Now suppose that $k = \mathbb{Q}(\sqrt{-d})$ has class number one, take $S = \{\mathcal{P}_1, \dots, \mathcal{P}_n\}$ and let $\pi = \pi_1 \pi_2 \cdots \pi_n$, where π_i generates $\mathcal{P}_i$ for each i . We claim that for

$$\mathcal{O}' = \left\{ \begin{bmatrix} x & \pi y \\ z/\pi & w \end{bmatrix} \mid x, y, z, w \in O_d \right\},$$

$\mathcal{O}'_{\mathcal{P}} = \mathcal{O}_{\mathcal{P}}$ for $\mathcal{P} \notin S$, and $\mathcal{O}'_{\mathcal{P}}$ stabilizes a vertex adjacent to the one stabilized by $\mathcal{O}_{\mathcal{P}}$ in $\mathcal{T}_{\mathcal{P}}$. If $\mathcal{P} \notin S$ then $\mathcal{O}_{\mathcal{P}}$ and $\mathcal{O}'_{\mathcal{P}}$ coincide because each π_i is a unit in $R_{\mathcal{P}}$ (by virtue of not belonging to $\mathcal{P}$). On the other hand, if $\mathcal{P} = \mathcal{P}_i$ belongs to S then $\pi = \pi_1 \cdots \pi_n$ generates the unique maximal ideal of $R_{\mathcal{P}_i}$, just like π_i , since π_j is a unit in $R_{\mathcal{P}_i}$ for $j \neq i$. In this case, taking $\{\mathbf{e}_1, \mathbf{e}_2\}$ to be a basis for the lattice $L \subset V$ stabilized by $\mathcal{O}_{\mathcal{P}}$, $L' = \text{span}\{\pi_i \mathbf{e}_1, \mathbf{e}_2\}$ has distance one from L in $\mathcal{T}_{\mathcal{P}}$, and direct computation shows that $\mathcal{O}'_{\mathcal{P}}$ stabilizes L' . This proves the claim. The associated Eichler order is:

$$\mathcal{E} = \mathcal{O} \cap \mathcal{O}' = \left\{ \begin{bmatrix} x & \pi y \\ z & w \end{bmatrix} \mid x, y, z, w \in O_d \right\}$$

In the notation of [13, Dfn. 6.6.6], $\mathcal{E}$ has *level* $\prod_{i=1}^n \mathcal{P}_i$. As established in the proof of Theorem 11.5.1 of [13], its projectivized group of units $P(\mathcal{E}^1)$ has index $\prod_{i=1}^n (n(\mathcal{P}_i) + 1)$ in $P(\mathcal{O}^1) = \text{PGL}_2(O_d)$, where for each i , $n(\mathcal{P}_i)$ is the norm of the prime ideal $\mathcal{P}_i$. (We will compute these in the next subsection.) That result bounds the index of $P(\mathcal{E}^1)$, which is our $\tilde{\Gamma}_{\pi}$, above by 2^n in $P(N(\mathcal{E}))$.

In fact equality holds in this case, as follows from results of [5]. Below we will adapt that work to explicitly exhibit elements of $P(N(\mathcal{E}))$ that are *odd* at prescribed $\mathcal{P}_i$. More precisely, we seek elements of A which, included in $A_{\mathcal{P}}$, fix the vertices of the tree $\mathcal{T}_{\mathcal{P}}$ corresponding to $\mathcal{O}_{\mathcal{P}}$ and $\mathcal{O}'_{\mathcal{P}}$ for all primes $\mathcal{P}$ except for prescribed $\mathcal{P}_i$, where these vertices are flipped. Each such element will belong to $P(N(\mathcal{E}))$.

We continue to take $S = \{\mathcal{P}_1, \dots, \mathcal{P}_n\}$ and $\pi = \pi_1 \cdots \pi_n$, where π_i generates $\mathcal{P}_i$ for each i . For any such i , upon including $M_{\pi} = \begin{bmatrix} 0 & \pi \\ 1 & 0 \end{bmatrix}$ in $M_2(k_{\mathcal{P}_i})$ it takes $L = \text{span}\{\mathbf{e}_1, \mathbf{e}_2\}$ to $L' = \text{span}\{\pi_i \mathbf{e}_1, \mathbf{e}_2\}$, and L' to the scaled copy $\pi_i L$ of L (cf. [13, p. 353, top half]). On the other hand, for a prime ideal $\mathcal{P} \notin S$, π is a unit in $R_{\mathcal{P}}$ and hence as $R_{\mathcal{P}}$ -modules, $L = L'$ is preserved by $M_{\pi} \in M_2(k_{\mathcal{P}})$. If $n = 1$ then $\Gamma_{\pi} = P(N(\mathcal{E})) = \langle \tilde{\Gamma}_{\pi}, M_{\pi} \rangle$ contains $\tilde{\Gamma}_{\pi} = P(\mathcal{E}^1)$ with index two.

Now assuming that $n > 1$, and fixing $i < n$, we may choose $k_i, l_i \in O_d$ such that

$$k_i \pi_1 \cdots \pi_i + l_i \pi_{i+1} \cdots \pi_n = 1$$

(using that the products $\pi_1 \cdots \pi_i$ and $\pi_{i+1} \cdots \pi_n$ are relatively prime), and define:

$$(6) \quad N_i = \begin{bmatrix} \pi_1 \cdots \pi_i & \pi \\ l_i & -k_i \pi_1 \cdots \pi_i \end{bmatrix}.$$

We claim that N_i is odd at $\mathcal{P}_j$ if and only if $j \leq i$. Exploiting the property of k_i and l_i , we find that

$$(7) \quad k_i \pi_1 \cdots \pi_i N_i(\mathbf{e}_1) + l_i N_i(\mathbf{e}_2) = \pi_1 \cdots \pi_i \mathbf{e}_1$$

$$(8) \quad -\pi_{i+1} \cdots \pi_n N_i(\mathbf{e}_1) + N_i(\mathbf{e}_2) = -\mathbf{e}_2,$$

Now take $\mathcal{P} = \mathcal{P}_j$ for some $j \leq i$, and let $L = \text{span}(\mathbf{e}_1, \mathbf{e}_2)$ and $L' = \text{span}(\pi_j \mathbf{e}_1, \mathbf{e}_2)$ for a generator π_j of $\mathcal{P}_j$, $R_{\mathcal{P}}$ -modules at distance one in the tree for $\text{SL}_2(k_{\mathcal{P}})$. Both $N_i(\mathbf{e}_1)$ and $N_i(\mathbf{e}_2)$ belong to L' —in fact, $N_i(\mathbf{e}_2) \in \pi_j L \subset L'$ —so $N_i(L) \subseteq L'$, and

the equations above imply that equality holds. Rewriting (7), and multiplying both sides of (8) by $\pi_1 \cdots \pi_i$, we obtain

$$\begin{aligned} k_i N_i(\pi_1 \cdots \pi_i \mathbf{e}_1) + l_i N_i(\mathbf{e}_2) &= \pi_1 \cdots \pi_i \mathbf{e}_1 \\ -\pi_{i+1} \cdots \pi_n N_i(\pi_1 \cdots \pi_i \mathbf{e}_1) + \pi_1 \cdots \pi_i N_i(\mathbf{e}_2) &= -\pi_1 \cdots \pi_i \mathbf{e}_2, \end{aligned}$$

Since $N_i(L') \subseteq \pi_j L$, this shows that $N_i(L') = \pi_j L$, hence that N_i is odd at $\mathcal{P}_j$.

For any prime ideal $\mathcal{P}$ such that $\mathcal{P} \neq \mathcal{P}_j$ for any $j \leq i$, upon including N_i in $M_2(k_{\mathcal{P}})$, equations (7) and (8) imply that N_i sends the $R_{\mathcal{P}}$ -lattice $L = \text{span}\{\mathbf{e}_1, \mathbf{e}_2\}$ to itself, since $\pi_1 \cdots \pi_i$ is a unit in $R_{\mathcal{P}}$. This proves the claim that N_i is odd at $\mathcal{P}$ if and only if $\mathcal{P} = \mathcal{P}_j$ for some $j \leq i$.

Finally, taking $\mathcal{P} = \mathcal{P}_j$ for some $i < j \leq n$, multiplying both sides of (7) by a generator π_j for $\mathcal{P}_j$ and rearranging terms slightly in (8) yields:

$$\begin{aligned} k_i \pi_1 \cdots \pi_i N_i(\pi_j \mathbf{e}_1) + l_i \pi_j N_i(\mathbf{e}_2) &= \pi_1 \cdots \pi_i (\pi_j \mathbf{e}_1) \\ -\pi_{i+1} \cdots \pi_n \pi_j N_i(\pi_j \mathbf{e}_1) + N_i(\mathbf{e}_2) &= -\mathbf{e}_2, \end{aligned}$$

Since $\pi_1 \cdots \pi_i$ is a unit in $R_{\mathcal{P}_j}$, in this case we have $N_i(L') = L'$. We have shown that for each $\mathcal{P} = \mathcal{P}_j \in S$, upon including N_i in $M_2(k_{\mathcal{P}})$ it stabilizes the edge of $\mathcal{T}_{\mathcal{P}}$ joining the vertices corresponding to L and L' . It flips this edge if $j \leq i$ and otherwise fixes it pointwise. For all such j , this implies that $P(N_i)$ normalizes $P(\mathcal{E}_{\mathcal{P}_j})$ since $P(\mathcal{E}_{\mathcal{P}_j}^1)$ is the subgroup of $\text{PGL}_2(k_{\mathcal{P}})$ fixing this edge. In the case that $\mathcal{P} \notin S$ we showed above that upon including N_i in $M_2(k_{\mathcal{P}})$ it stabilizes the vertex of $\mathcal{T}_{\mathcal{P}}$ corresponding to L . In this case $\mathcal{O}_{\mathcal{P}} = \mathcal{O}_{\mathcal{P}}$ by construction, so $P(\mathcal{E}_{\mathcal{P}}^1) = P(\mathcal{O}_{\mathcal{P}}^1)$ is the vertex stabilizer in question and hence contains $P(N_i)$. Therefore Lemma 11.4.1 of [13] implies that N_i normalizes $P(\mathcal{E}^1)$.

There is a homomorphism $\phi: P(N(\mathcal{E})) \rightarrow (\mathbb{Z}/2\mathbb{Z})^n$ with $\phi(M)$ non-zero in the j th component if and only if M is odd at $\mathcal{P}_j$. The kernel of ϕ is then $P(\mathcal{E}^1)$, $\phi(M_{\pi}) = (1, \dots, 1)$, and for each $i < n$, $\phi(N_i)$ has non-zero j th entry if and only if $j \leq i$. Therefore ϕ surjects $(\mathbb{Z}/2\mathbb{Z})^n$.

To summarize, applying the machinery of [3], [5] and [13] to our case we obtain:

Proposition 3.1. *For a square-free positive integer d such that $\mathbb{Q}(\sqrt{-d})$ has class number one, and distinct primes $\pi_1, \dots, \pi_n \in \mathcal{O}_d$, taking $\pi = \pi_1 \cdots \pi_n$,*

$$\tilde{\Gamma}_{\pi} = \left\{ \begin{bmatrix} x & \pi y \\ z & w \end{bmatrix} \in \text{PGL}_2(\mathcal{O}_d) \mid x, y, z, w \in \mathcal{O}_d \right\}$$

has index $\prod_{i=1}^n (n(\mathcal{P}_i) + 1)$ in $\text{PGL}_2(\mathcal{O}_d)$, where for each i , $n(\mathcal{P}_i)$ is the norm of the prime ideal generated by π_i . For $M_{\pi} = \begin{bmatrix} 0 & \pi \\ 1 & 0 \end{bmatrix}$ and $N_i = \begin{bmatrix} \pi_1 \cdots \pi_i & \pi \\ l_i & -k_i \pi_1 \cdots \pi_i \end{bmatrix}$, $1 \leq i < n$, where for each such i , $k_i \pi_1 \cdots \pi_i + l_i \pi_{i+1} \cdots \pi_n = 1$:

$$\Gamma_{\pi} = \langle \tilde{\Gamma}_{\pi}, M_{\pi}, N_1, \dots, N_{n-1} \rangle$$

is a maximal orientation-preserving discrete group with $\Gamma_{\pi} \cap \text{PGL}_2(\mathcal{O}_d) = \tilde{\Gamma}_{\pi}$. It normalizes $\tilde{\Gamma}_{\pi}$, with $[\Gamma_{\pi} : \tilde{\Gamma}_{\pi}] = 2^n$. Every maximal, orientation-preserving discrete group commensurable with but not equal to $\text{PGL}_2(\mathcal{O}_d)$ is conjugate to Γ_{π} for some $\pi \in \mathcal{O}_d$ as above.

Remark. Since every discrete group of isometries is contained in a maximal discrete group, every orientable orbifold commensurable with $Q_0 \doteq \mathbb{H}^3 / \text{PGL}_2(\mathcal{O}_d)$ covers Q_0 or some Q_{π} .

Definition 3.2. For distinct primes $\pi_1, \dots, \pi_n \in O_d$ and $\pi = \pi_1 \cdots \pi_n$, taking $\tilde{\Gamma}_\pi$ and Γ_π as in Proposition 3.1, we let $\tilde{Q}_\pi = \mathbb{H}^3/\tilde{\Gamma}_\pi$ and $Q_\pi = \mathbb{H}^3/\Gamma_\pi$. We call $\tilde{Q}_\pi$ the *canonical cover* of Q_π .

Proposition 3.1 implies that each Q_π as defined above is a *minimal orientable orbifold*; that is, it does not cover any other orientable orbifold.

Example 3.3. Taking $d = 3$, the two smallest primes in O_3 are $\sqrt{-3}$ and 2, with norms 3 and 4, respectively. (More on this in Section 3.2.) In applying the formulas of Proposition 3.1 to $\pi = 2\sqrt{-3}$ with $\pi_1 = \sqrt{-3}$ and $\pi_2 = 2$, we can then set $k_1 = 1$ and $l_1 = \frac{1}{2}(1 - \sqrt{-3})$, yielding

$$N_1 = \begin{bmatrix} \sqrt{-3} & 2\sqrt{-3} \\ \frac{1-\sqrt{-3}}{2} & -\sqrt{-3} \end{bmatrix}$$

We note that although our choice of k_1 and l_1 yielded a class N_1 having order two in $\mathrm{PGL}_2(\mathbb{Q}(\sqrt{-3}))$, there are infinitely many possible such choices which would not have, and even for this choice, $M_\pi N_1 \neq N_1 M_\pi$. It is generally true though that any choice of k_i and l_i yields an N_i with the property that $N_i^2 \in \tilde{\Gamma}_\pi$, and $[M_\pi, N_i] \in \tilde{\Gamma}_\pi$ (cf. the Remark after Theorem 3.4).

3.2. Norms of primes in O_3 . Here, with an eye toward [8], we give an explicit list of all minimal orientable orbifolds in the commensurability class $\mathcal{C}_3$ of $\mathbb{H}^3/\mathrm{PGL}_2(O_3)$ that have volume at most $2v_0$, where $v_0 \approx 1.0149$ is the volume of a regular ideal tetrahedron in $\mathbb{H}^3$. We are not aware of such a list already in the literature.

Theorem 3.4. For $\pi \in O_3$, let $\tilde{\Gamma}_\pi = \left\{ \begin{bmatrix} x & \pi y \\ z & w \end{bmatrix} \in \mathrm{PGL}_2(O_3) \mid x, y, z, w \in O_3 \right\}$.

The minimal orientable orbifolds of volume at most $2v_0$ that are commensurable to $Q_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_3)$ are Q_0 itself, which has volume $\frac{v_0}{12}$, and the $Q_\pi \doteq \mathbb{H}^3/\Gamma_\pi$ below, where for the following prime π , $\Gamma_\pi = \langle \tilde{\Gamma}_\pi, M_\pi \rangle < \mathrm{PGL}_2(\mathbb{Q}(\sqrt{-3}))$ for $M_\pi = \begin{bmatrix} 0 & \pi \\ 1 & 0 \end{bmatrix}$:

- $\pi = \sqrt{-3}$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 4$ and $\mathrm{vol}(Q_\pi) = 2\frac{v_0}{12}$;
- $\pi = 2$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 5$ and $\mathrm{vol}(Q_\pi) = \frac{5}{2}\frac{v_0}{12}$;
- $\pi = 2 \pm \sqrt{-3}$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 8$ and $\mathrm{vol}(Q_\pi) = 4\frac{v_0}{12}$;
- $\pi = 1 \pm 2\sqrt{-3}$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 14$ and $\mathrm{vol}(Q_\pi) = 7\frac{v_0}{12}$;
- $\pi = 4 \pm \sqrt{-3}$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 20$ and $\mathrm{vol}(Q_\pi) = 10\frac{v_0}{12}$;
- $\pi = 5$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 26$ and $\mathrm{vol}(Q_\pi) = 13\frac{v_0}{12}$;
- $\pi = 2 \pm 3\sqrt{-3}$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 32$ and $\mathrm{vol}(Q_\pi) = 16\frac{v_0}{12}$;
- $\pi = 5 \pm 2\sqrt{-3}$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 38$ and $\mathrm{vol}(Q_\pi) = 19\frac{v_0}{12}$;
- $\pi = 4 \pm 3\sqrt{-3}$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 44$ and $\mathrm{vol}(Q_\pi) = 22\frac{v_0}{12}$;

for the following composite $\pi = \pi_1\pi_2$, $\Gamma_\pi = \langle \tilde{\Gamma}_\pi, M_\pi, N_1 \rangle$ where $M_\pi = \begin{bmatrix} 0 & \pi \\ 1 & 0 \end{bmatrix}$ as above and $N_1 = \begin{bmatrix} \pi_1 & \pi \\ l & -k\pi_1 \end{bmatrix}$ for $k, l \in O_3$ such that $k\pi_1 + l\pi_2 = 1$:

- $\pi = 2\sqrt{-3}$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 20$ and $\mathrm{vol}(Q_\pi) = 5\frac{v_0}{12}$;
- $\pi = \sqrt{-3}(2 \pm \sqrt{-3})$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 32$ and $\mathrm{vol}(Q_\pi) = 8\frac{v_0}{12}$;
- $\pi = 2(2 \pm \sqrt{-3})$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 40$ and $\mathrm{vol}(Q_\pi) = 10\frac{v_0}{12}$;
- $\pi = \sqrt{-3}(1 \pm 2\sqrt{-3})$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 56$ and $\mathrm{vol}(Q_\pi) = 14\frac{v_0}{12}$;
- $\pi = 7$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 64$ and $\mathrm{vol}(Q_\pi) = 16\frac{v_0}{12}$;
- $\pi = 2(1 \pm 2\sqrt{-3})$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 70$ and $\mathrm{vol}(Q_\pi) = \frac{35}{2}\frac{v_0}{12}$;

• $\pi = \sqrt{-3}(4 \pm \sqrt{-3})$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 80$ and $\mathrm{vol}(Q_\pi) = 20 \frac{v_0}{12}$;
 and for $\pi = 2\sqrt{-3}(2 \pm \sqrt{-3})$, with $[\mathrm{PGL}_2(O_3) : \tilde{\Gamma}_\pi] = 160$, $\Gamma_\pi = \langle \tilde{\Gamma}_\pi, M_\pi, N_1, N_2 \rangle$
 and $\mathrm{vol}(Q_\pi) = 20 \frac{v_0}{12}$, where $N_1 = \begin{bmatrix} \sqrt{-3} & \pi \\ 1 & 3 \pm 2\sqrt{-3} \end{bmatrix}$ and $N_2 = \begin{bmatrix} 2\sqrt{-3} & \pi \\ -1 & -3 \mp \sqrt{-3} \end{bmatrix}$.

Remark. For a product $\pi = \pi_1 \pi_2$ of two primes as in the Theorem, we record some computational observations. First, $N_1 = \begin{bmatrix} \pi_2 & \pi_1 \\ -k & l \end{bmatrix} \begin{bmatrix} 0 & \pi_1 \\ 1 & 0 \end{bmatrix}$ is the product of a member of $\tilde{\Gamma}_{\pi_1}$ with M_{π_1} . Second, $N_1 M_\pi = \pi_1 \begin{bmatrix} \pi_2 & \pi \\ -k & l \pi_2 \end{bmatrix}$ is $\mathrm{PGL}_2(\mathbb{Q}(\sqrt{-3}))$ -equivalent to an analogous product of a member of $\tilde{\Gamma}_{\pi_2}$ with M_{π_2} . Third, $N_1 M_\pi N_1^{-1} M_\pi^{-1} \in \tilde{\Gamma}_\pi$. Finally, $N_1^2 = \pi_1 \begin{bmatrix} \pi_1 + l \pi_2 & (1-k)\pi \\ l(1-k) & l \pi_2 + k^2 \pi_1 \end{bmatrix}$ belongs to $\tilde{\Gamma}_\pi$ up to $\mathrm{PGL}_2(\mathbb{Q}(\sqrt{-3}))$ -equivalence.

Among other things, these combine to show that Γ_π is an extension of $\tilde{\Gamma}_\pi$ by $(\mathbb{Z}/2\mathbb{Z})^2$, as predicted by the general theory recorded in Proposition 3.1.

With an eye to producing Theorem 3.4's list, we now describe the prime ideals of the ring of integers O_3 of the number field $k = \mathbb{Q}(\sqrt{-3})$. For every prime ideal $\mathcal{P}$ in O_3 , $\mathcal{P} \cap \mathbb{Z} = p\mathbb{Z}$ is generated by some integer prime p ; we say that $\mathcal{P}$ lies over p . Because k is a quadratic extension of $\mathbb{Q}$, there are exactly three possibilities for the ideal pO_3 in O_3 generated by p :

- $pO_3 = \mathcal{P}^2$ (we say p ramifies in k);
- $pO_3 = \mathcal{P}\mathcal{P}'$ for another prime ideal $\mathcal{P}' \subset O_3$ (p decomposes); or
- $pO_3 = \mathcal{P}$ is itself prime (p is inert).

Here since O_3 is a principal ideal domain, the product $\mathcal{P}_1 \mathcal{P}_2$ of ideals $\mathcal{P}_1$ and $\mathcal{P}_2$ is the set $\{r \pi_1 \pi_2 \mid r \in O_3\}$, where π_i generates $\mathcal{P}_i$ for $i = 1, 2$. In general, we will denote the ideal generated by $\pi \in O_3$ as (π) or πO_3 . Below we classify the integer primes less than 48 according to the categories above for O_3 .

- *ramifies*: $3 = -(\sqrt{-3})^2$ only (among all integer primes; see [13, Th. 0.3.8])
- *decomposes*: 7, 13, 19, 31, 37, 43
- *inert*: 2, 5, 11, 17, 23, 29, 41, 47

This can be checked using [13, Lemma 0.3.10]. Furthermore, the decomposing primes decompose as follows: $7 = (2 - \sqrt{-3})(2 + \sqrt{-3})$, $13 = (1 - 2\sqrt{-3})(1 + 2\sqrt{-3})$, $19 = (4 - \sqrt{-3})(4 + \sqrt{-3})$, $31 = (2 + 3\sqrt{-3})(2 - 3\sqrt{-3})$, $37 = (5 + 2\sqrt{-3})(5 - 2\sqrt{-3})$, and $43 = (4 + 3\sqrt{-3})(4 - 3\sqrt{-3})$. For each of these p , each listed factor of p generates an ideal factor of pO_3 as described above.

The *norm* of the prime ideal $\mathcal{P}$ is the cardinality $n_{\mathcal{P}}$ of the field $O_3/\mathcal{P}$; here also the square $|\pi|^2$ of the complex modulus of a generator π for $\mathcal{P}$. If $\mathcal{P}$ lies over $p \in \mathbb{N}$ then $n_{\mathcal{P}} = p$ if p ramifies or decomposes, and $n_{\mathcal{P}} = p^2$ if p is inert. Thus of all primes of O_3 , only the inert integer primes 2 and 5, plus $\sqrt{-3}$ and each of those listed above that decomposes an integer prime, have norm less than 48.

Proof of Theorem 3.4. The fact that $Q_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_3)$ itself has volume $v_0/12$ comes from the fact that its fundamental domain is one twelfth of a regular ideal tetrahedron. For $\pi = \pi_1 \cdots \pi_n \in O_3$, a product of distinct primes, by Proposition 3.1 the minimal orbifold Q_π thus has volume $2^{-n} \left[\prod_{i=1}^n (n(\mathcal{P}_i) + 1) \right] \frac{v_0}{12}$, where $\mathcal{P}_i = \pi_i O_d$ for each i . If $\mathcal{P} = \pi O_d$ is prime, Q_π thus has volume at most $2v_0$ if and only if $n(\mathcal{P}) \leq 47$. Using the prior computations of this section to list the norms of primes of O_3 that are at most 47, in non-decreasing order, yields:

$$3, 4, 7, 13, 19, 25, 31, 37, 43$$

The nine primes or pairs of primes with these norms yield the first list of Theorem 3.4. For a product $\mathcal{P} = \mathcal{P}_1\mathcal{P}_2$ of two primes such that Q_π has volume at most $2v_0$, the product $(n(\mathcal{P}_1) + 1)(n(\mathcal{P}_2) + 1)$ must not exceed 96. Since the smaller of the two norms is at least three, the larger can be at most 23. The seven pairs of norms having a small enough product yield the second list of the Theorem. For products of three primes, only the three smallest primes yield a product less than $2^3 \cdot 24 = 192$. The product corresponding to the four distinct primes with smallest norm, $4 \cdot 5 \cdot 8 \cdot 8 = 1280$ is already larger than $2^4 \cdot 24 = 384$. This plus a short induction argument imply that there are no relevant minimal orbifolds associated to a product of four or more primes. $\square$

3.3. Cusps of minimal orbifolds and their covers. The main result of [5] implies that each minimal orbifold Q_π of Definition 3.2 has one cusp. Here we describe how the cusps of its canonical cover $\tilde{Q}_\pi$ project to it and to the other minimal orbifold $Q_0 = \mathbb{H}^2/\mathrm{PGL}_2(O_d)$ that $\tilde{Q}_\pi$ is guaranteed to cover. Then, in Corollary 3.6 we classify the cusps of $\tilde{Q}_\pi$ up to isomorphism.

Here a *cusp* of an orbifold $N = \mathbb{H}^3/\Gamma$ is the image of a horoball $B \subset \mathbb{H}^3$ with the property that its projection to N factors through an isomorphism of its quotient by the stabilizer of its ideal point in Γ . The cusp is determined up to isomorphism by the quotient of ∂B , a Euclidean 2-orbifold, so we name the type of the cusp after this orbifold. (This is all standard, see eg. [10, §2].)

Proposition 3.5. *For $d \in \mathbb{N}$ square-free such that $\mathbb{Q}(\sqrt{-d})$ has class number one, and a product $\pi = \pi_1 \cdots \pi_n$ of distinct primes in O_d , the orbifold $\tilde{Q}_\pi$ of Definition 3.2 has 2^n cusps, each projecting isomorphically to Q_π . Each monomial summand of the expansion of $\prod_{i=1}^n (n(\mathcal{P}_i) + 1)$ is the degree of the projection of a cusp of $\tilde{Q}_\pi$ to $Q_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_d)$, where each $n(\mathcal{P}_i)$ is the norm of the ideal generated by π_i .*

Proof. As observed in Lemma 3.1 of [5], any group commensurable to $\mathrm{PGL}_2(O_d)$ has $\mathbb{Q}(\sqrt{-d})$ as its collection of parabolic fixed points (called “cusps” in [5]). This holds true in particular for $\tilde{\Gamma}_\pi$. Now for any $\bar{\epsilon} = (\epsilon_1, \dots, \epsilon_n) \in \{0, 1\}^n$, define $\pi^{\bar{\epsilon}} = \pi_1^{\epsilon_1} \cdots \pi_n^{\epsilon_n}$. We claim that for any $\bar{\epsilon}' \neq \bar{\epsilon}$, $\pi^{\bar{\epsilon}'}$ is not $\tilde{\Gamma}_\pi$ -equivalent to $\pi^{\bar{\epsilon}}$.

Suppose there was an element $A = \begin{bmatrix} x & \pi y \\ z & w \end{bmatrix}$ of $\tilde{\Gamma}_\pi$, for $x, y, z, w \in O_d$, taking $\pi^{\bar{\epsilon}}$ to $\pi^{\bar{\epsilon}'}$ for some $\bar{\epsilon}' \neq \bar{\epsilon}$. Then by definition we would have:

$$x\pi^{\bar{\epsilon}} + \pi y = (z\pi^{\bar{\epsilon}} + w)\pi^{\bar{\epsilon}'}$$

As $\bar{\epsilon} \neq \bar{\epsilon}'$, at least one of the following is true: $\pi^{\bar{\epsilon}}$ has some π_j as a factor that $\pi^{\bar{\epsilon}'}$ does not, or vice-versa. In the former case, solving the equation above for $w\pi^{\bar{\epsilon}'}$ shows that π_j must divide w ; in the latter, solving for $x\pi^{\bar{\epsilon}}$ shows that π_j divides x . In either case, it then follows that π_j divides the determinant of A , contradicting that A belongs to $\mathrm{PGL}_2(O_d)$. This proves the claim.

The claim implies that $\tilde{Q}_\pi = \mathbb{H}^3/\tilde{\Gamma}_\pi$ has at least 2^n cusps, since there are 2^n distinct $\pi^{\bar{\epsilon}}$. But since $\tilde{Q}_\pi$ covers Q_π , which has one cusp by [5], with degree 2^n , it has exactly 2^n cusps and each projects isomorphically to the cusp of Q_π .

We now address the cusps' covers to Q_0 . For $\bar{\epsilon} \in \{0, 1\}^n$, conjugating the lower-triangular matrices by $\begin{bmatrix} 1 & \pi^{\bar{\epsilon}} \\ 0 & 1 \end{bmatrix}$, we find that the stabilizer of $\pi^{\bar{\epsilon}}$ in $\mathrm{PGL}_2(O_d)$ consists of elements of the form

$$\begin{bmatrix} a + \pi^{\bar{\epsilon}}c & \pi^{\bar{\epsilon}}(d - a - \pi^{\bar{\epsilon}}c) \\ c & d - \pi^{\bar{\epsilon}}c \end{bmatrix}$$

for $a, c, d \in O_d$ such that a and d are units. For a complete set $c_1, \dots, c_k$ of coset representatives for the ideal generated by $\pi/\pi^{\bar{\epsilon}}$ in O_d , we claim that the collection $M_1, \dots, M_k$ is a complete collection of left-coset representatives for the stabilizer of $\pi^{\bar{\epsilon}}$ in $\tilde{\Gamma}_\pi$ in the corresponding subgroup of $\mathrm{PGL}_2(O_d)$, where for each i ,

$$M_i = \begin{bmatrix} 1 + \pi^{\bar{\epsilon}} c_i & -(\pi^{\bar{\epsilon}})^2 c_i \\ c_i & 1 - \pi^{\bar{\epsilon}} c_i \end{bmatrix}.$$

To check this we first observe that $M_i^{-1} M_j$ does not lie in $\tilde{\Gamma}_\pi$ for any $i \neq j$, since π does not divide its upper right entry $(\pi^{\bar{\epsilon}})^3(c_i - c_j)$ by the choice of the c_i . Therefore the M_i represent pairwise distinct cosets of the stabilizer of $\pi^{\bar{\epsilon}}$ in $\tilde{\Gamma}_\pi$ inside the corresponding subgroup of $\mathrm{PGL}_2(O_d)$. It follows that the cusp of $\tilde{Q}_\pi$ corresponding to $\pi^{\bar{\epsilon}}$ covers the corresponding cusp of Q_0 with degree at least k , where k is the number of the coset representatives c_i for $\pi/\pi^{\bar{\epsilon}}$ in O_d .

What is k ? For $\bar{\epsilon} = (\epsilon_1, \dots, \epsilon_n) \in \{0, 1\}^n$, the definition of $\pi^{\bar{\epsilon}}$ gives that

$$\pi/\pi^{\bar{\epsilon}} = \pi_{i_1} \pi_{i_2} \cdots \pi_{i_m}$$

for some $m \leq n$, where $i_1 < i_2 < \dots < i_m$ is the sequence of indices such that $\epsilon_{i_j} = 0$. Since $n(\mathcal{P}_{i_j})$ is the index in O_d of the prime ideal $\mathcal{P}_{i_j}$ generated by π_{i_j} for each i , the index k of the prime ideal generated by $\pi/\pi^{\bar{\epsilon}}$ thus equals $\prod_{j=1}^m n(\mathcal{P}_{i_j})$ by the Chinese remainder theorem.

We thus have a lower bound, for each of the 2^n cusps of $\tilde{Q}_\pi$, of the degree of its cover to the cusp of Q_0 . As the collection of cusps corresponds bijectively to the collection of $\bar{\epsilon} \in \{0, 1\}^n$, the covering degree bounds sum to

$$\sum_{\bar{\epsilon} \in \{0, 1\}^n} \prod_{j=1}^{m_{\bar{\epsilon}}} n(\mathcal{P}_{i_j}),$$

where for each $\bar{\epsilon} \in \{0, 1\}^n$ the sequence $i_1 < \dots < i_{m_{\bar{\epsilon}}}$ records those indices at which 0 is the entry of $\bar{\epsilon}$. But this sum is exactly the covering degree $\prod_{i=1}^n (n(\mathcal{P}_i) + 1)$, so each of the given lower bounds is an equality. $\square$

Corollary 3.6. *For a positive square-free integer d such that $\mathbb{Q}(\sqrt{-d})$ has class number one, and a product π of distinct primes in O_d , all cusps of $\tilde{Q}_\pi$ have the same type as the cusp of $Q_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_d)$, as does the sole cusp of Q_π .*

Proof. Proposition 3.5 implies that the covering transformation group $\Gamma_\pi/\tilde{\Gamma}_\pi \cong (\mathbb{Z}/2\mathbb{Z})^n$ of $\tilde{Q}_\pi$ over Q_π acts simply transitively (by isometries) on the collection of cusps of $\tilde{Q}_\pi$, each of which hence projects isometrically to the cusp of Q_π . Since one of these cusps also projects isometrically to the cusp of Q_0 , all cusps of $\tilde{Q}_\pi$, hence also of Q_π have the same type as the cusp of Q_0 . $\square$

The next result really just records two special cases of the proof of Proposition 3.5.

Corollary 3.7. *For $d \in \mathbb{N}$ square-free such that $\mathbb{Q}(\sqrt{-d})$ has class number one, and a product $\pi = \pi_1 \cdots \pi_n$ of distinct primes in O_d , for $\tilde{\Gamma}_\pi$ defined as in Proposition 3.1, the stabilizers of 0 in $\tilde{\Gamma}_\pi$ and $\mathrm{PGL}_2(O_d)$ are equal, whereas the stabilizer of ∞ in $\tilde{\Gamma}_\pi$ has index $\prod_{i=1}^n n(\mathcal{P}_i)$ in the corresponding subgroup of $\mathrm{PGL}_2(O_d)$.*

Proof. We note that ∞ is $\tilde{\Gamma}_\pi$ -equivalent to 1, via $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, and 0 to π via $\begin{pmatrix} 1 & \pi \\ 0 & 1 \end{pmatrix}$. The corresponding cases of the proof of Proposition 3.1 are $\bar{\epsilon} = (0, \dots, 0)$, for which $\pi^{\bar{\epsilon}} = 1$, and $\bar{\epsilon} = 1$, for which $\pi^{\bar{\epsilon}} = \pi$. From the Proposition's proof, the stabilizer of

1 in $\tilde{\Gamma}$ has index $\prod_{i=1}^n n(\mathcal{P}_i)$ in the stabilizer in $\mathrm{PGL}_2(O_d)$, and the corresponding index for the stabilizers of π is 1, giving the result.

More directly, noting that the stabilizers of 0 and ∞ in $\tilde{\Gamma}_\pi$ are respectively its lower- and upper-triangular subgroups, it follows from the definition of $\tilde{\Gamma}_\pi$ that it contains the entire stabilizer of 0 in $\mathrm{PGL}_2(O_d)$. And on the other hand, it is straightforward to check that for a set $\{c_1, \dots, c_k\}$ of distinct coset representatives for the ideal $\mathcal{P}$ generated by π in O_d , the set $\left\{ \begin{bmatrix} 1 & c_1 \\ 0 & 1 \end{bmatrix}, \dots, \begin{bmatrix} 1 & c_k \\ 0 & 1 \end{bmatrix} \right\}$ consists of distinct right coset representatives for the stabilizer of ∞ in $\tilde{\Gamma}_\pi$, inside the same stabilizer in $\mathrm{PGL}_2(O_d)$. This gives an alternate take on the proof. $\square$

Example 3.8. We now restrict to the cases $d = 1$ and $d = 3$ of interest here, first observing that $R_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_1)$ has a $(2, 4, 4)$ -turnover cusp, and that $Q_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_3)$ has a $(2, 3, 6)$ -turnover cusp. In fact, we claim for each of $d = 1$ and 3 , the generators ρ_f and ρ_e for $\mathrm{PGL}_2(O_d)$ respectively described in (1) and (2) comprise a generating set for its stabilizer of ∞ .

To see this, recall from Definition 2.1 the fundamental domain T_d for the action of $\mathrm{PGL}_2(O_d)$ on $\mathbb{H}^3$ described there intersects the horoball B above the plane at height 1 in $\Delta_d \times [1, \infty)$, where Δ_d is the triangle with vertices at 0 and $\frac{1}{2}(1 \pm i)$ or $\frac{1}{2}(1 \pm 1/\sqrt{-3})$ in the respective cases. Thus $\Delta_d \times \{1\}$ is a fundamental domain for the stabilizer of ∞ in $\mathrm{PGL}_2(O_d)$, with ρ_f identifying its two sides containing 0 and ρ_e identifying the opposite side to itself. This proves the claim, and the geometry of Δ_d determines the type of the cusp.

Now suppose that π is a product of distinct primes in O_d for $d = 1$ or 3 . By Corollary 3.6, every cusp of $\tilde{Q}_\pi$ has the same type as that of R_0 or Q_0 , respectively. Let Λ_π be the stabilizer of ∞ in $\tilde{\Gamma}_\pi$ from Proposition 3.1. Note that ρ_f , being diagonal, belongs to Λ_π . We claim that it generates Λ_π , together with

$$\rho_\pi \doteq \begin{bmatrix} -1 & \pi \\ 0 & 1 \end{bmatrix},$$

and that the triangle Δ_π similar to Δ_d (for $d = 1$ or 3) that has 0 at one vertex and $\pi/2$ at the midpoint of the opposite edge is a fundamental domain for the action of Λ_π on $\mathbb{C}$. Note that ρ_π defined as above is an order-two rotation fixing $\pi/2 \in \mathbb{C}$. It therefore identifies the edge Δ_π that is opposite 0 to itself. As ρ_f identifies the two edges of Δ_π containing 0, Δ_π is a fundamental domain for the action of $\langle \rho_f, \rho_\pi \rangle$.

By Proposition 3.7, the stabilizer of ∞ in $\tilde{\Gamma}$ has index $\prod_{i=1}^n n(\mathcal{P}_i)$ in the corresponding subgroup of $\mathrm{PGL}_2(O_d)$. The claim above will follow by observing that this is also the ratio of the area of Δ_π to that of Δ_d in each case. In the case $d = 1$, Δ_π has a right angle at 0 and hence area $\frac{|\pi|^2}{4}$, where $|\pi|$ is the complex modulus of π . In the case $d = 3$, Δ_π is equilateral and has area $\frac{|\pi|^2}{4\sqrt{3}}$. In turn, $|\pi|^2$ is the product $|\pi_1|^2 \cdots |\pi_n|^2$, where π is the product of primes $\pi_1, \dots, \pi_n$. As we will discuss in the next subsection, $|\pi_i|^2 = n(\mathcal{P}_i)$ for each i , so $|\pi|^2 = \prod_{i=1}^n n(\mathcal{P}_i)$. Computing the areas of Δ_1 and Δ_3 as $1/4$ and $1/(4\sqrt{3})$ from their descriptions in Definition 2.1, we obtain the claim.

Again looking ahead to [8], we record a final result on covers of Q_0 and the Q_π .

Proposition 3.9. *For $d \in \mathbb{N}$ square-free such that $\mathbb{Q}(\sqrt{-d})$ has class number one, and a product π of n distinct primes of O_d , let $Q_\pi = \mathbb{H}^3/\Gamma_\pi$ and $\tilde{Q}_\pi = \mathbb{H}^3/\tilde{\Gamma}_\pi$ as in Definition 3.2. For any cover $P \rightarrow Q_\pi$, represented by a subgroup Γ of $\mathbb{H}^3/\Gamma_\pi$, let its canonical cover be $\tilde{P} = \mathbb{H}^3/\tilde{\Gamma}$, where $\tilde{\Gamma} = \Gamma \cap \mathrm{PGL}_2(O_d)$. Then:*

- $\tilde{P}$ also covers $\tilde{Q}_\pi$.
- The cover $\tilde{P} \rightarrow P$ is normal with covering group in $(\mathbb{Z}/2\mathbb{Z})^n$, and it does not factor through any non-trivial morphism between covers of $\tilde{Q}_\pi$.

Furthermore, each covering transformation of $\tilde{P} \rightarrow P$ preserves the partition of the cusps of $\tilde{P}$ according to which cusp of $\tilde{Q}_\pi$ they cover, with only the identity taking any partition class to itself.

$$\begin{array}{ccc}
 \tilde{P} = \mathbb{H}^3/\tilde{\Gamma} & & \\
 \downarrow & \searrow & \\
 \tilde{Q}_\pi = \mathbb{H}^3/\tilde{\Gamma}_\pi & & P = \mathbb{H}^3/\Gamma \\
 \downarrow & \searrow & \downarrow \\
 Q = \mathbb{H}^3/\mathrm{PGL}_2(O_d) & & Q_\pi = \mathbb{H}^3/\Gamma_\pi
 \end{array}$$

Proof. For Γ and $\tilde{\Gamma}$ as above, note that $\tilde{\Gamma}$ is contained in $\tilde{\Gamma}_\pi = \Gamma_\pi \cap \mathrm{PGL}_2(O_d)$, so $\tilde{P} = \mathbb{H}^3/\tilde{\Gamma}$ covers $\tilde{Q}_\pi$ as well as P . In fact $\tilde{\Gamma} = \Gamma \cap \tilde{\Gamma}_\pi$, since $\Gamma < \Gamma_\pi$, so $\tilde{P} \rightarrow P$ does not factor through any other cover of $\tilde{Q}_\pi$. And since Γ_π normalizes $\tilde{\Gamma}_\pi$, its subgroup Γ normalizes $\tilde{\Gamma} = \Gamma \cap \tilde{\Gamma}_\pi$ by the diamond isomorphism theorem, with the quotient $\Gamma/\tilde{\Gamma}$ isomorphic to $\Gamma\tilde{\Gamma}_\pi/\tilde{\Gamma}_\pi$, itself a subgroup of $\Gamma_\pi/\tilde{\Gamma}_\pi \cong (\mathbb{Z}/2\mathbb{Z})^n$. This is the covering group of $\tilde{P} \rightarrow P$.

The cusps of $\tilde{Q}_\pi$ correspond to $\tilde{\Gamma}_\pi$ -orbits of points of $\mathbb{Q}(\sqrt{-d})$ (cf. the proof of Prop. 3.5), and the cusps of $\tilde{P}$ to $\tilde{\Gamma}$ -orbits. Since Γ normalizes $\tilde{\Gamma}_\pi$, for any $\gamma \in \Gamma$, if $x, y \in \mathbb{Q}(\sqrt{-d})$ are $\tilde{\Gamma}_\pi$ -equivalent then so are $\gamma.x$ and $\gamma.y$:

$$y = \tilde{\gamma}.x, \tilde{\gamma} \in \tilde{\Gamma}_\pi \quad \Rightarrow \quad \gamma.y = \gamma\tilde{\gamma}.x = (\gamma\tilde{\gamma}\gamma^{-1})\gamma.x$$

Thus the covering transformation of $\tilde{P}$ induced by γ takes the cusps of $\tilde{P}$ corresponding to x and y , both of which cover the cusp of $\tilde{Q}_\pi$ corresponding to x , to cusps covering the cusp of $\tilde{Q}_\pi$ corresponding to $\gamma.x$. The fact that only the identity takes any partition class to itself follows from the consequence of Proposition 3.5 recorded directly below it—that the covering transformation group of $\tilde{Q}_\pi \rightarrow Q_\pi$ acts simply transitively on the cusps of $\tilde{Q}_\pi$. $\square$

4. PROOF OF THEOREM 1.1

We give the proof of Theorem 1.1 in this section. It goes as follows: first, in Lemma 4.1 below, we establish the Theorem's first assertion, that any orbifold of volume less than $v_7/2$ is commensurable to one of $\mathbb{H}^3/\mathrm{PGL}_2(O_1)$ or $\mathbb{H}^3/\mathrm{PGL}_2(O_3)$. As incommensurable orbifolds are in particular non-isometric, we may separately

handle the two commensurability classes for the isometry classification. Sections 4.2 and 4.3 respectively accomplish this for the commensurability classes above.

Lemma 4.1. *Let $v_7 \approx 0.44445$ be the volume of $\mathbb{H}^3/\mathrm{PGL}_2(O_7)$. A complete, non-compact arithmetic hyperbolic 3-orbifold with volume less than $v_7/2 \approx 0.222225$ is commensurable to exactly one of $Q_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_3)$ or $R_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_1)$. (Here for $d = 1, 3, 7$, O_d is the ring of integers of $\mathbb{Q}(\sqrt{-d})$.)*

Proof. We will show that the three commensurability classes of non-compact, arithmetic hyperbolic 3-orbifolds having the minimal orientable members of smallest volume are:

- (1) That of $\mathbb{H}^3/\mathrm{PGL}_2(O_3)$, with volume approximately 0.084578.
- (2) That of $\mathbb{H}^3/\mathrm{PGL}_2(O_1)$, with volume approximately 0.15266
- (3) That of $\mathbb{H}^3/\mathrm{PGL}_2(O_7)$, with volume approximately 0.44445.

Since every non-orientable orbifold is double covered by an orientable orbifold, this will establish the result.

The proof is essentially that of [13, Th. 11.7.2], plus an extra restriction that holds in the present special case of non-compact orbifolds. As recorded at the top of [13, p. 367], for an arithmetic manifold with field of definition an imaginary quadratic number field $k = \mathbb{Q}(\sqrt{-d})$, the minimum volume μ in its commensurability class is bounded below by $\frac{|\Delta_k|^{1/2}}{32}$, where Δ_k is the discriminant of k .

Because we are focused on *non-compact* orbifolds, however, a stronger bound holds than the above. This is because the quaternion algebra A associated to any non-compact orbifold is the matrix algebra $M_2(k)$, which does not ramify at any finite place. Thus in the estimate about halfway down p. 366 of [13]:

$$\mu \geq \frac{|\Delta_k|^{3/2}}{8\pi^2} \frac{\zeta_k(2)}{h_k} \prod_{\mathcal{P}|\Delta(A)} \left(\frac{N(\mathcal{P}) - 1}{2} \right) = \frac{|\Delta_k|^{3/2}}{8\pi^2} \frac{\zeta_k(2)}{h_k}.$$

Here h_k is the class number of k and $\Delta(A)$ is the *discriminant* of the quaternion algebra A , the product of the primes determining places at which A ramifies (cf. [13, p. 100]). As noted above, this collection of primes is empty when A is the matrix algebra, yielding the right-hand equality above. Further applying the inequality (11.30) from [13] we obtain

$$\mu \geq \frac{|\Delta_k|^{1/2}}{8\pi^2} \frac{1}{h_k} \Delta_k \zeta_k(2) \geq \frac{|\Delta_k|^{1/2}}{8\pi^2} \frac{1}{h_k} \frac{h_k}{2w} (2\pi)^2 = \frac{|\Delta_k|^{1/2}}{4w}.$$

The first inequality above is just a rearrangement of the one above it, and the second is obtained by substituting using [13, (11.30)]. There, w is the order of the group of roots of unity in O_d , and we have used the observation from [13] that the regulator R appearing there takes the value 1 for quadratic imaginary fields. After simplifying, we obtain the final equality.

For any positive square free integer d such that $|\Delta_k| \geq 13$ for $k = \mathbb{Q}(\sqrt{-d})$, the group of units in O_d has order two, as observed in [13]. For such d we therefore obtain $\mu \geq \sqrt{13}/8 > 0.45$. For the remaining d , we can explicitly compute the volume using the formula $\mu = \frac{|\Delta_k|^{3/2} \zeta_k(2)}{8\pi^2}$ from [13]. We recall that for $k = \mathbb{Q}(\sqrt{-d})$, where d is a square-free integer, $\Delta_k = -d$ if $-d \equiv 1 \pmod{4}$, and $\Delta_k = -4d$ otherwise. Using this to augment the table of approximate ζ_k -values in [13, p.367], we obtain:

The result follows from these values and the bounds for fields k with $|\Delta_k| \geq 13$. $\square$

Field k	$ \Delta_k $	$\zeta_k(2)$	μ
$\mathbb{Q}(\sqrt{-1})$	4	1.50670301	0.15266
$\mathbb{Q}(\sqrt{-2})$	8	1.75141751	0.50192
$\mathbb{Q}(\sqrt{-3})$	3	1.28519096	0.08457
$\mathbb{Q}(\sqrt{-7})$	7	1.89484145	0.44445
$\mathbb{Q}(\sqrt{-11})$	11	1.49613186	0.69130

4.1. Generalities on non-orientable orbifolds. We will understand the non-orientable orbifolds in $\mathcal{C}_1$ and $\mathcal{C}_3$ as quotients of certain orientable orbifolds.

Definition 4.2. An orientable orbifold is *achiral* if it has an orientation-reversing self-isometry; otherwise it is *chiral*.

It is a basic fact that every non-orientable orbifold $O = \mathbb{H}^3/\Gamma$ has a degree-two cover by the achiral orientable orbifold $O_0 = \mathbb{H}^3/\Gamma_0$, where $\Gamma_0 = \Gamma \cap \mathrm{PSL}_2(\mathbb{C})$, being the quotient of O_0 by an orientation-reversing involution.

Fact 4.3. A minimal orientable orbifold covers at most one non-orientable orbifold.

This holds because the orientation-preserving isometry group of any orientable orbifold has index at most two in its full isometry group, but if the orientable orbifold is minimal then its orientation-preserving isometry group must be trivial (since otherwise it would cover another orientable orbifold, cf. [19, Prop. 13.2.1]).

Lemma 4.4. For a square-free positive integer d such that $\mathbb{Q}(\sqrt{-d})$ has class number one, $\mathbb{H}^3/\mathrm{PGL}_2(O_d)$ is achiral. For a prime $\pi \in O_d$, the associated minimal orbifold Q_π is achiral if and only if the integer prime p over (π) is inert or ramifies. If p splits, there is an orientation-reversing isometry $Q_\pi \rightarrow Q_{\bar{\pi}}$.

Proof. Conjugation by the reflection r of $\mathbb{H}^3$ extending $z \mapsto \bar{z}$ preserves $\mathrm{PGL}_2(O_d)$, since it takes $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{PSL}_2(\mathbb{C})$ to $\begin{pmatrix} \bar{a} & \bar{b} \\ \bar{c} & \bar{d} \end{pmatrix}$. For a prime $\pi \in O_d$, it takes the groups $\tilde{\Gamma}_\pi$ and Γ_π of Proposition 3.1 to $\tilde{\Gamma}_{\bar{\pi}}$ and $\Gamma_{\bar{\pi}}$, respectively, by their explicit descriptions in the Proposition. Therefore r induces an orientation-reversing self-isometry of $\mathbb{H}^3/\mathrm{PGL}_2(O_d)$, and isometries from Q_π to $Q_{\bar{\pi}}$ for each such π .

If $p = \pi\bar{\pi}$ splits then (π) and $(\bar{\pi})$ are distinct ideals, hence Q_π and $Q_{\bar{\pi}}$ are distinct. In the case that $p = \pi$ is inert, $Q_{\bar{\pi}} = Q_\pi$ and hence r induces a self-isometry of Q_π . In the ramified case we can use the fact that $p = \pi\bar{\pi} \in (\pi)^2$ to find $u \in O_d$ such that

$$\pi\bar{\pi} = \pi^2 u \quad \Rightarrow \quad \bar{\pi} = \pi u.$$

Therefore u is a unit, so $\begin{pmatrix} 0 & \bar{\pi} \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} u & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & \pi \\ 1 & 0 \end{pmatrix}$, and again $Q_\pi = Q_{\bar{\pi}}$.

In the inert or ramified case, the involution induced by r is the unique orientation-reversing isometry by Fact 4.3. In the split case, there is no orientation-reversing self-isometry of Q_π : if there was one, composing the isometry induced by r with it would supply an orientation-preserving isometry to $Q_{\bar{\pi}}$, a contradiction. $\square$

We now directly consider the sole product of primes relevant to this paper.

Example 4.5. For π equal to the product of primes $i\sqrt{3}$ and 2 in O_3 , we have $\Gamma_\pi = \langle \tilde{\Gamma}_\pi, N_1, M_\pi \rangle$, where in this case we can take

$$N_1 = \begin{pmatrix} i\sqrt{3} & 2i\sqrt{3} \\ 2 & -i\sqrt{3} \end{pmatrix} \quad \text{and} \quad M_\pi = \begin{pmatrix} 0 & \pi \\ 1 & 0 \end{pmatrix}.$$

Entrywise complex conjugation takes each of N_1 and M_π to its product with $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \in \tilde{\Gamma}_\pi$, hence induces an orientation-reversing self-isometry of Q_π . Note that in this case $i\sqrt{3}$ ramifies 3, and 2 is inert, over O_3 .

On the other hand, a *non-minimal* orientable orbifold may be the orientable double cover of more than one non-orientable orbifold. The special case addressed below suffices to consider for this paper.

Lemma 4.6. *If O is a complete, orientable, hyperbolic 3-orbifold with self-isometry group isomorphic to a Klein four-group with elements μ and ρ that are respectively orientation-preserving and -reversing, then the orbifolds $O/\langle\rho\rangle$ and $O/\langle\rho\mu\rangle$ are non-isometric.*

Proof. Write $O = \mathbb{H}^3/\Gamma$ for a discrete group $\Gamma < \mathrm{PSL}_2(\mathbb{C})$. The isometries μ and ρ of O are induced by elements of the normalizer $N(\Gamma)$ of Γ , which we will refer to by the same names, and the quotient orbifolds $O/\langle\rho\rangle$ and $O/\langle\rho\mu\rangle$ are the quotients of $\mathbb{H}^3$ by the groups $\langle\Gamma, \rho\rangle$ and $\langle\Gamma, \rho\mu\rangle$.

A putative isometry g from one quotient orbifold to the other would be induced by an isometry of $\mathbb{H}^3$ that conjugates one group to the other. Because Γ is the intersection of each of these groups with $\mathrm{PSL}_2(\mathbb{C})$, which is normal in the full isometry group of $\mathbb{H}^3$, the conjugating isometry g would also belong to $N(\Gamma)$. It would conjugate ρ to $\rho\mu$ in the Klein four-group $N(\Gamma)/\Gamma$, which is impossible since they are distinct elements of this abelian group. $\square$

4.2. Orbifolds commensurable with $\mathbb{H}^3/\mathrm{PGL}_2(O_1)$. We next establish the classification of low-volume orbifolds commensurable with $\mathbb{H}^3/\mathrm{PGL}_2(O_1)$. We begin with the *minimal* orientable orbifolds.

Lemma 4.7. *The two minimal orientable orbifolds in the commensurability class $\mathcal{C}_1$ of $R_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_1)$ that have volume less than v_7 are R_0 —with volume $v_1/6$, where $v_1 \approx 0.91596559$ is Catalan’s constant—and the orbifold R_{1+i} in $\mathcal{C}_1$ associated to the prime $(1+i)O_1$ in the sense of Proposition 3.1, with volume $v_1/4$.*

Proof. The only prime in O_1 with norm less than 5 is $(1+i)O_1$, which ramifies $2 = (1+i)(1-i)$ since $1-i = 2-(1+i)$ belongs to $(1+i)O_1$. (The integer prime 3 is inert in O_1 , hence with norm 9, and in general a prime over $p \in \mathbb{Z}$ has norm at least p ; cf. Section 3.2.) The minimal orbifold R_π associated to a prime π with norm p has volume $(p+1)/2 \cdot v$ by Proposition 3.1, where v is the volume of $\mathbb{H}^3/\mathrm{PGL}_2(O_1)$. Thus R_π has volume at least $3v$ for any prime $\pi \neq 1+i$. An inductive argument using Proposition 3.1 shows that $\mathrm{vol}(R_\pi)$ is still larger if π is a product of more than one prime.

As mentioned in Section 2.1, $\mathrm{PGL}_2(O_1)$ is the orientation-preserving symmetry group of a tiling $\mathcal{T}_1$ of $\mathbb{H}^3$ by regular ideal octahedra. The full symmetry group of a regular ideal octahedron has order 48, so the orientation-preserving symmetry group has order 24. The volume of $\mathbb{H}^3/\mathrm{PGL}_2(O_1)$ is thus 1/24th that of the regular ideal octahedron, or, since this divides into four ideal tetrahedra with dihedral angles $\frac{\pi}{2}$, $\frac{\pi}{4}$, and $\frac{\pi}{4}$, one-sixth of that tetrahedron’s volume v_1 . It is well known that $v_1 \approx 0.91596559$ is Catalan’s constant (see eg. [19, §7.2]).

Because $(1+i)O_1$ has norm 2, R_{1+i} has volume $\frac{3}{2} \frac{v_1}{6} = \frac{v_1}{4}$. For any other prime or product of primes π , R_π has volume at least $3 \frac{v_1}{6} = \frac{v_1}{2} > v_7$ by the proof’s first paragraph. $\square$

Proposition 4.8. *At most nine members of the commensurability class $\mathcal{C}_1$ of $R_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_1)$ have volume less than $v_7/2$. In order of increasing volume, they are:*

- *With volume $v_1/12$, where $v_1 \approx 0.91596559$ is Catalan's constant, the volume of an ideal tetrahedron in $\mathbb{H}^3$ with dihedral angles $\frac{\pi}{2}$, $\frac{\pi}{4}$, and $\frac{\pi}{4}$: the unique non-orientable orbifold $\bar{R}_0$ double-covered by R_0 .*
- *With volume $\frac{v_1}{8}$: the unique non-orientable orbifold $\bar{R}_{1+i}$ double-covered by the minimal orbifold R_{1+i} in $\mathcal{C}_1$ associated to the prime $(1+i)O_1$.*
- *With volume $\frac{v_1}{6}$: R_0 , or one of six non-orientable orbifolds double-covered by the three double covers of R_0 —two each.*

Proof. Of the two orientable minimal orbifolds in $\mathcal{C}_1$ with volume less than v_7 , the volume $v_1/4 \approx 0.22899$ of R_{1+i} is greater than $v_7/2 \approx 0.2222$. While $\mathrm{vol}(R_0) = v_1/6 < v_7/2$, twice the volume of R_0 is $v_1/3$, greater than $v_7/2$. Since every orientable orbifold in $\mathcal{C}_1$ covers a minimal orientable orbifold, the only orientable orbifold in $\mathcal{C}_1$ having volume less than $v_7/2$ is R_0 .

Every non-orientable orbifold $\bar{P}$ in $\mathcal{C}_1$ has an orientable cover P of degree 2 with $\mathrm{vol}(P) = 2\mathrm{vol}(\bar{P})$. Therefore if $\bar{P}$ has volume less than $v_7/2$ then P has volume less than v_7 , so by the paragraph above, the minimal orientable orbifolds covered by P can consist only of R_0 and/or R_{1+i} . Even stronger, since volume multiplies by covering degree, P can cover R_0 with degree at most two, or R_{1+i} with degree at most one. By Fact 4.3 and Lemma 4.4—noting that $1+i$ ramifies 2 in O_1 — R_0 and R_{1+i} each cover unique non-orientable orbifolds.

It remains only to enumerate the non-orientable orbifolds covered by degree-two covers of R_0 . We recall from Section 2.2 that there are three degree-two covers of R_0 , corresponding to the stabilizers of 0 under the following right-permutation representations of $\mathrm{PGL}_2(O_1)$:

- (1) $\rho_f \mapsto [0, 1], \rho_e \mapsto [1, 0], \rho_v \mapsto [1, 0]$
- (2) $\rho_f \mapsto [1, 0], \rho_e \mapsto [0, 1], \rho_v \mapsto [0, 1]$
- (3) $\rho_f \mapsto [1, 0], \rho_e \mapsto [1, 0], \rho_v \mapsto [1, 0]$

By Lemma 2.4, we may take $T_1 \cup \rho_v(T_1)$ as a fundamental domain for the subgroups of (1) and (3), and $T_1 \cup \rho_f(T_1)$ as a fundamental domain for subgroup (2). Applying the Lemma again, we obtain the following generating sets for the three subgroups:

- (1) $\{\rho_f, \rho_e \rho_v\}$
- (2) $\{\rho_f^2, \rho_e, \rho_f \rho_e \rho_f^{-1}, \rho_v, \rho_f \rho_v \rho_f^{-1}\}$
- (3) $\{\rho_f \rho_v, \rho_v \rho_f, \rho_e \rho_v\}$

One can check directly that each of these subgroups is normalized by the reflection r in the plane $\mathcal{H}$ over $\mathbb{R}$, using the fact (recorded in Section 2.1) that r conjugates ρ_f to ρ_f^{-1} and commutes with each of ρ_e and ρ_v . Therefore r induces an orientation-reversing involution of each cover.

For any of these three covers $P \rightarrow R_0$, as the cover has degree two it is normal with a single non-trivial covering transformation i (induced eg. by ρ_v in cases (1) and (3), and ρ_f in case (2)). The group of covering transformations is therefore a Klein four-group generated by i and r , with ir as a second orientation-reversing isometry. Thus $P/\langle r \rangle$ and $P/\langle ir \rangle$ are non-orientable orbifolds covered by P . $\square$

Proposition 4.9. *The nine orbifolds of Proposition 4.8 are pairwise non-isometric.*

Proof. As orbifolds with different volumes are non-isometric, and an orientable orbifold is not isometric to a non-orientable orbifold, what we need to do is distinguish the six non-orientable orbifolds of volume $v_1/6$ from each other. By Lemma 4.6, it is enough to show that their orientable double covers are distinct up to isometry, since isometric non-orientable orbifolds have isometric orientable double covers.

Using the descriptions (1), (2), and (3) from the proof of Proposition 4.8, we can distinguish cover (2) from the other two by its cusp. From Section 2.1, the subgroup of $\mathrm{PGL}_2(O_1)$ fixing ∞ is a $(2, 4, 4)$ -rotation group generated by ρ_f and ρ_e , with ρ_f and $\rho_f\rho_e$ representing its conjugacy classes of elements of order 4. As neither ρ_f nor $\rho_f\rho_e$ belongs to the stabilizer of 0 under representation (2), the stabilizer of ∞ in this subgroup must not have any element of order 4. But the corresponding stabilizers under (1) and (3) respectively contain ρ_f and $\rho_f\rho_e$.

We can distinguish (1) from (3) by features of their singular loci, which may be discerned from the fundamental domain $(T_1 \cup \rho_v(T_1))$ in both cases) and face-pairings on it. Since the face-pairings of (3) are rotations in the edges of intersection between T_1 and $\rho_v(T_1)$, this group is the rotation group in T_1 , and the corresponding orbifold's singular locus is the one-skeleton. In particular, its vertices are the three non-ideal vertices of T_1 . On the other hand, the singular locus Σ in case (1) has only one vertex, as two vertices of T_1 are identified by face-pairings and the third lies in the interior of a geodesic component of Σ . $\square$

4.3. Orbifolds commensurable with $\mathbb{H}^3/\mathrm{PGL}_2(O_3)$. Here we perform the analog for the commensurability class $\mathcal{C}_3$ of $Q_0 = \mathbb{H}^3/\mathrm{PGL}_2(O_3)$ as we did for $\mathcal{C}_1$ the previous subsection.

Proposition 4.10. *Exactly 19 members of the commensurability class $\mathcal{C}_3$ have volume less than $v_7/2$. In order of increasing volume, the orientable members are:*

- *With volume $\frac{v_0}{12}$, where $v_0 \approx 1.01494146$ is the volume of a regular ideal tetrahedron in $\mathbb{H}^3$: Q_0 .*
- *With volume $v_0/6$: the unique double cover O^2 of Q_0 , and the minimal orientable orbifold $Q_{\sqrt{-3}} = \mathbb{H}^3/\Gamma_{\sqrt{-3}}$ in $\mathcal{C}_3$ associated to the prime $\sqrt{-3}O_3$.*
- *With volume $5v_0/24$: the minimal orientable orbifold $Q_2 = \mathbb{H}^3/\Gamma_2$ in $\mathcal{C}_3$ associated to the prime $2O_3$.*

Each of Q_0 , $Q_{\sqrt{-3}}$, and Q_2 double-covers a unique non-orientable orbifold having half its volume, denoted as $\bar{Q}_0$, $\bar{Q}_{\sqrt{-3}}$, and $\bar{Q}_2$, respectively. There are exactly two non-orientable orbifolds double-covered by O^2 . In order of increasing volume, the remaining non-orientable members of $\mathcal{C}_3$ are:

- *With volume $v_0/8$: the unique non-orientable orbifold $\bar{O}^3$ double covered by the unique degree-3 cover O^3 of Q_0 .*
- *With volume $v_0/6$: exactly two orbifolds double-covered by the unique degree-4 cover O^4 of Q_0 .*
- *With volume $5v_0/24$: a total of six distinct orbifolds respectively double-covered by the double covers O^5 , P_2^1 , and P_2^2 of Q_2 , where O^5 also covers Q_0 ; and the unique orbifold $\bar{Q}_{2\sqrt{-3}}$ double covered by the minimal orientable orbifold $Q_{2\sqrt{-3}}$ associated to the product of primes $2\sqrt{-3}O_3$.*

Proof. Like in the previous subsection, the full isometry classification will largely follow from the classification of orientable orbifolds, plus the description of their isometry groups. We embark on that first. Again, the first step is to list all minimal

orientable orbifolds with volume less than v_7 . Appealing to Theorem 3.4, we obtain the orbifolds below, which we pair with their volumes:

$$(Q_0, v_0/12), (Q_{\sqrt{-3}}, v_0/6), (Q_2, 5v_0/24), (Q_{2\pm\sqrt{-3}}, v_0/3), (Q_{2\sqrt{-3}}, 5v_0/12).$$

Recall that $v_7 \approx 0.44445$, so since $v_0 \approx 1.01494146$ we have $5v_0/12 < v_7 < v_0/2$ and this list is complete. We recall that the orientable minimal orbifold $Q_{\sqrt{-3}} = \mathbb{H}^3/\Gamma_3$ and its cover by O^4 , were described in Lemma 2.7; and that the orientable minimal orbifold $Q_2 = \mathbb{H}^3/\Gamma_2$, of volume $5v_0/24$, and its cover by O^5 , were described in Lemma 2.8.

As every orientable orbifold covers a minimal orientable orbifold, the next step is to enumerate the orientable orbifolds of volume at most v_7 that cover at least one of Q_0 , $Q_{\sqrt{-3}}$, and Q_2 , the other minimal orbifolds having volume greater than $v_7/2$. This was handled in Section 2. We recall the relevant results:

- The covers O^i of Q_0 of degree $i \in \{2, 3, 4, 5\}$ are listed in Definition 2.3.
- The degree-two covers of each of $Q_{\sqrt{-3}}$ and Q_2 that are not (obviously) among the O^i are described in Lemma 2.10.

By the volume observations of the previous paragraph, we have now listed every orientable orbifold in $\mathcal{C}_3$ having volume at most v_7 .

We now classify these orientable orbifolds, proceeding in order of increasing volume. The two with volume $v_0/6$, O^2 and $Q_{\sqrt{-3}}$, are non-isometric. This can be seen from the fact that $Q_{\sqrt{-3}}$ is minimal and O^2 is not, or by considering their cusps: it follows from Corollary 3.6 and Lemma 2.5, respectively, that $Q_{\sqrt{-3}}$ has a $(2, 3, 6)$ -cusp and O^2 a $(3, 3, 3)$ -cusp.

The five with volume $v_0/3$ are $P_{\sqrt{-3}}^1$, $P_{\sqrt{-3}}^2$, O^4 , and the minimal orbifolds $Q_{2\pm\sqrt{-3}}$. It follows from Lemma 2.9 that $P_{\sqrt{-3}}^1$ and $P_{\sqrt{-3}}^2$ are orientation-reversing isometric, and from Lemma 2.10 that they are *not* orientation-preserving isometric: if so, composing the OR isometry with the inverse of an OP isometry would give an OR self-isometry. This likewise holds for $Q_{2\pm\sqrt{-3}}$ by Lemma 4.4. The three isometry classes are now distinguished by their cusps: each $P_{\sqrt{-3}}^i$ has a single $(3, 3, 3)$ -cusp; O^4 has two $(2, 3, 6)$ -cusps; and each of $Q_{2\pm\sqrt{-3}}$ has a single $(2, 3, 6)$ -cusp.

Finally, there are four with volume $5v_0/12$: the three covers O^5 , P_2^1 , and P_2^2 of Q_2 corresponding to the subgroups listed in Lemma 2.9, plus the minimal orbifold $Q_{2\sqrt{-3}}$. The latter has a single $(2, 3, 6)$ -cusp, by Corollary 3.6, whereas $O^5 = \tilde{Q}_2$ has two $(2, 3, 6)$ -cusps, and by Lemma 2.11, P_2^i has one $(3, 3, 3)$ -cusp for each $i \in \{1, 2\}$. The singular loci distinguish these two P_2^i , as described in the Lemma. This completes the classification of orientable orbifolds in $\mathcal{C}_3$ with volume less than v_7 , up to isometry.

Of the orbifolds listed above, those with volume less than $v_7/2$ are Q_0 , O^2 , $Q_{\sqrt{-3}}$, and Q_2 . To complete the classification of orbifolds in $\mathcal{C}_3$ having volume less than $v_7/2$, we classify the non-orientable orbifolds double-covered by each orientable orbifold of volume less than v_7 . We distinguish by the number of orbifolds covered.

- Those orientable orbifolds that double-cover no non-orientable orbifold: $P_{\sqrt{-3}}^1$ and $P_{\sqrt{-3}}^2$, by Lemma 2.10, and $Q_{2\pm\sqrt{-3}}$, by Lemma 4.4.
- Those covering a unique non-orientable orbifold: the minimal orbifolds Q_0 , $Q_{\sqrt{-3}}$, and Q_2 , by Lemma 4.4 and Fact 4.3; $Q_{2\sqrt{-3}}$ by Example 4.5; and O^3 , by Lemma 2.6.

- Those double-covering exactly two non-orientable orbifolds: O^2 , O^4 , O^5 , P_2^1 , and P_2^2 ; all applications of Lemma 4.6.

Toward the final bullet above, we note that each of the orbifolds mentioned there has a unique orientation-preserving involution, the covering involution of its cover to Q_0 (in the case of O^2), $Q_{\sqrt{-3}}$ (O^4), or Q_2 (O^5 , P_2^1 , and P_2^2). In particular, the covers $O^5 \rightarrow Q_0$ and $O^4 \rightarrow Q_0$ are not normal, as can be readily showed, and $O^4 \rightarrow Q_0$ does not factor through O^2 . The latter can be seen for instance from the fact that O^4 has $(2, 3, 6)$ -cusps and O^2 a single $(3, 3, 3)$ -cusp, consequences of the descriptions in Lemmas 2.7 and 2.5. $\square$

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